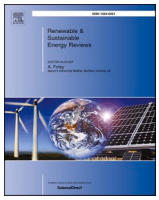
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Progress and perspectives in source-sink matching for CCUS: A critical review

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ABSTRACT

As an essential technology for significant emission reductions, carbon capture, utilization, and storage (CCUS) rely critically on source-sink matching. This fundamental element plays a vital role by connecting CO_2 emission points (sources) to appropriate geological storage sites (sinks) efficiently and cost-effectively. Such matching is indispensable for realizing CCUS's potential in widespread deployment and environmental benefits. Although the matching technology framework is relatively mature, comprehensive reviews specifically addressing source-sink matching remain limited. This paper systematically reviews the current advancements in CCUS source-sink matching research, focusing on the three core dimensions of "source-sink-matching," and synthesizes key elements, including the spatiotemporal distribution patterns of global CO_2 emission sources, geological capacity assessment, and potential distribution, as well as transportation modes connecting sources and sinks. Through in-depth analysis, this paper identifies four systemic challenges: disconnects in technology-economy-policy systems; discrepancies in data accuracy and scale between sources and sinks; inflexible transport optimization methods and insufficient cluster collaboration; and algorithm-physical mechanism disconnections lacking dynamic evolution characterization. Building on a thorough examination of current challenges and developments, this study proposes strategic recommendations for enhancing CCUS source-sink matching frameworks. These recommendations aim to accelerate global CCUS deployment planning and advance international carbon neutrality goals.

1. Introduction

Human activities continue to push greenhouse gas (GHG) emissions closer to critical Earth system tipping points, exacerbating the climate crisis [1]. The IPCC's Sixth Assessment Report underscores that only immediate and profound reductions in emissions can mitigate catastrophic climate impacts [2]. Recent data (Fig. 1) indicate that fossil fuel and industrial CO_2 emissions reached an all-time high of 36.3 Gt in 2024 – a 0.9 % year-over-year increase [3]. Concurrently, global mean surface temperatures for 2011–2020 were 1.1 °C above pre-industrial levels (1850–1900) [4]. This accelerated warming is driving unprecedented glacier retreat, sea-level rise, and extreme weather frequency, threatening global ecosystem stability and socioeconomic security [5]. Consequently, addressing emissions from major industrial point sources (e.g., cement kilns, coal-fired power plants, and petroleum refineries) has become a critical scientific and technological challenge for stabilizing global temperatures [6–8].

As an integrated technological approach, carbon capture, utilization, and storage (CCUS) aims to establish a complete carbon management chain from emission sources to geological storage, thereby preventing industrial carbon emissions from entering the atmosphere [9–11]. The core of this technological system lies in the precise coupling of three key stages (Fig. 2): source capture (efficiently separating and concentrating point source CO_2 using technologies such as oxy-fuel combustion and post-combustion capture) [12–14], safe transportation (enabling large-scale carbon transport through high-pressure pipeline networks, cryogenic vessels, or regional hubs) [15–17], and permanent storage (injecting supercritical CO_2 into depleted oil/gas reservoirs, deep saline aquifers, or basalt formations, achieving geological isolation over millennium timescales through physical trapping and mineralization reactions) [18–20]. Climate modeling studies consistently identify CCUS as an indispensable technological pillar for achieving the Paris Agreement's 1.5 °C target. Model projections clearly demonstrate that emission reduction scenarios without CCUS fail to stabilize atmospheric

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CO_2-equivalent concentrations below the safe threshold of 450 ppm by 2100, underscoring its critical role in global deep decarbonization strategies [21].

CCUS projects have undergone commercial deployment worldwide (Fig. 3) [22–24]. Several major economies including China, the US, the EU, Australia, and Japan have implemented storage verification projects across various scales [25,26]. Global verified geological CO_2 storage data in 2023 show that the US (8061.812 Gt), China (3077.431 Gt), and Australia (502.43 Gt) rank top three, collectively accounting for 89.8 % [27]. Other countries advance industrialization through divergent technical pathways: Pioneering the world's first industrial-scale cross-border CO_2 storage solution, Norway's Northern Lights initiative utilizes depleted hydrocarbon reservoirs in the North Sea's Sleipner area. This hub offers 1.5 million tons of annual storage capacity, connecting major German and French industrial emitters through maritime transport of liquefied CO_2 [28]. Japan conducts a large-scale subsea bed storage demonstration in Tomakomai, verifying the feasibility of storage in pyroclastic rock formations [29]. Canada's Boundary Dam power station has significantly reduced carbon capture costs by implementing post-combustion capture technology [30].

Widespread implementation of CCUS infrastructure critically relies on the strategic configuration of source-sink matching systems [31,32]. This technical process requires coordinated spatial integration, capacity alignment, and dynamic feedback mechanisms between industrial CO_2 sources and geological sinks [33]. However, existing research primarily focuses on static factors like emission source cluster distribution characteristics and geological condition assessments of storage bodies, forming a planning framework dominated by geographical proximity. For instance, Lin et al. [34] conducted a study on carbon source-sink matching within China's steel sector, methodically assessing geographic spread of emission sources, storage capacities, and transportation network designs across diverse capture targets. This revealed the spatial mismatch between eastern industrial zones and western storage belts, and indicated that transportation costs gradually become the dominant factor as capture intensity increases (30 %–90 %), recommending priority deployment of CCUS demonstration projects at steel plants near basins. Hasbollah et al. [35] quantitatively assessed the storage potential of the Malay Basin using United States Department of Energy (US-DOE) and Carbon Sequestration Leadership Forum (CSLF) methodologies, confirming significant storage advantages in its deep saline aquifers and providing critical geological foundations for regional carbon management policies. Yu et al. [36] analyzed injection well data in the Montney shale region to compare storage characteristics between the Debolt and Montney formations, proposing differentiated siting strategies to balance seismic risks and engineering feasibility. Current research generally exhibits limitations: Most studies focus on techno-economic analyses of single emission sources or isolated storage sites, but generally lack systematic reviews of cross-regional, multi--factor synergistic source-sink matching mechanisms.

Although substantial breakthroughs have been achieved in techno-economic analyses of individual technical components, systematic reviews of cross-regional, multi-factor synergistic source-sink matching mechanisms remain scarce. This study focuses on the core issues in this field by establishing a three-dimensional "source-sink-matching" analytical framework to systematically synthesize recent global advances in source-sink matching research. As illustrated in Fig. 4, our research encompasses key technical aspects, including the spatiotemporal characteristics of CO_2 emission sources, assessment of geological storage capacity alongside site selection criteria, transport modes, and multi-scale matching models, and conducts hierarchical analysis of existing literature across four scales: global, national, regional, and industrial. Through hierarchical analysis and critical synthesis of existing literature, this study thoroughly examines four core issues: synergy bottlenecks in techno-economic policies, data accuracy and scale mismatch, limitations in transport and matching mechanisms, and intelligent algorithms and data-driven approaches. These findings provide crucial support for enhancing the theoretical foundations of CCUS source-sink matching systems and optimizing integrated carbon management solutions, offering new impetus for scaling CCUS deployment and advancing global carbon neutrality.

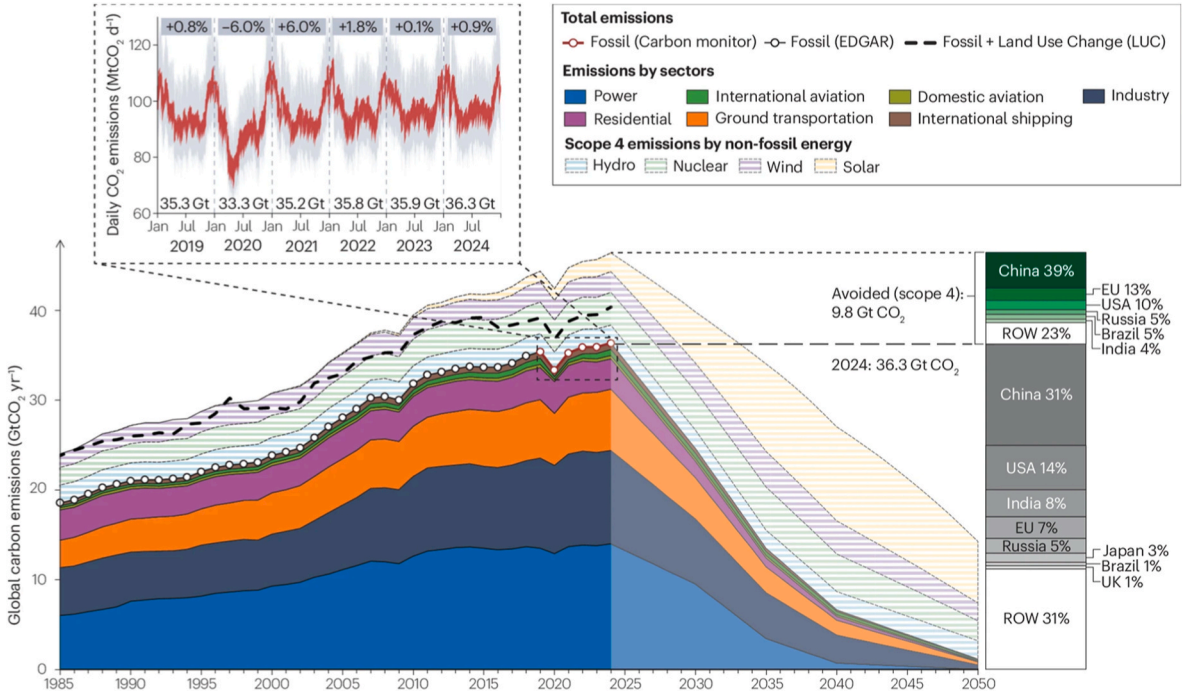


Fig. 1. Global CO_2 emission dynamics and decarbonization pathways (1985–2050). The main panel illustrates historical CO_2 outputs from fossil fuels and industrial processes (solid colors) alongside renewable energy-driven emission reductions (shaded colors), with an inset showing near-real-time CO_2 emission monitoring data post-2019 and year-on-year variations. The right-side bar chart displays national emission intensity (grayscale) and renewable energy mitigation equivalents (green) in 2024 [3].

2. Sources

2.1. Atmospheric CO_2

The greenhouse effect serves as a crucial mechanism for maintaining habitable temperatures in Earth's climate system [38]. Shortwave solar radiation penetrates the atmosphere and reaches at Earth's surface. This absorbed energy is subsequently re-emitted as longwave infrared radiation [39]. GHG selectively absorbs specific infrared wavelength bands through molecular vibrational modes, returning energy to the surface and atmosphere as thermal radiation, creating a "thermal blanket" effect. The greenhouse effect of CO_2 stems from the strong infrared absorption properties of its molecular structure: the vibrational frequency of its carbon-oxygen double bond corresponds closely to Earth's infrared emission spectrum, thereby enabling the efficient capture of thermally emitted energy from the surface.

Fig. 5 shows the composition proportions of major GHGs as: fluorinated gases (e.g. SF_6 and HFCs, 2 %), nitrous oxide (N_2O, 6 %), methane (CH_4, 16 %), and carbon dioxide (CO_2, 76 %) [40]. Table 1 quantifies climatic impacts using global warming potential (GWP), representing the equivalent warming effect of one ton of CO_2 over a century. Although CO_2 exhibits lower per-molecule radiative forcing than CH_4 (100-year GWP: $CO_2 = 1$, $CH_4 = 25$), its chemical inertness results in an atmospheric residence time of 200 years—significantly exceeding that of CH_4 (12 years) and that of N_2O (114 years) [41]. Furthermore, CO_2 emission volumes substantially exceed those of other GHGs, cementing its dominant role in the greenhouse effect [42].

2.2. CO_2 emissions

CO_2 emission sources exhibit significant heterogeneity, primarily categorized as anthropogenic and natural sources. Anthropogenic emissions constitute the dominant force behind contemporary climate change, driving global temperatures to 1.0 °C beyond pre-industrial baselines. Meeting the Paris Agreement's 1.5 °C target necessitates achieving net-zero emissions by mid-century and removing 100–1000 Gt of CO_2 from the atmosphere before 2100 [40]. Compared to unmanageable natural sources (e.g., wildfires), anthropogenic emission present greater opportunities for carbon capture, stationary point sources (e.g., power plants) are more amenable to capture technologies than mobile sources (e.g., transportation emissions) [45,46]. Notably, fossil fuel combustion processes—a major anthropogenic emission pathway—typically yield output containing high nitrogen content alongside CO_2, with significant CO_2 concentration fluctuations [47,48].

According to International Energy Agency (IEA) statistics, worldwide CO_2 emissions rose 0.8 % on an annual basis to 37.8 gigatons in 2024, marking an unprecedented peak. This surge elevated atmospheric CO_2 concentration to 422.5 ppm – a 3 ppm rise from 2023 levels and 50 % above pre-industrial baselines [49]. As shown in Fig. 6 (a), emissions from China, the US, India, the EU, and Japan collectively exceed 62.1 % of the global total [50]. China has consistently ranked first in global carbon emissions since 2005, with 2024 emissions reaching 12.6 Gt, where electricity/heat supply and industrial manufacturing constitute core domains of fossil energy consumption and carbon emissions [51]. Meanwhile, Japan, India, the EU, and the US recorded total CO_2 emissions of 1 Gt, 3 Gt, 2.4 Gt, and 4.5 Gt, respectively, in 2024. During the 2000–2024 period, developed economies are represented by the EU and the US. while energy-related CO_2 emissions from developed economies declined by 2.2 % in 2024, emissions from developing economies represented by China and India continued rising, highlighting the urgency of deep emission reductions. Fig. 6(b) displays the per capita CO_2 emissions of these countries. Although the US, Japan, and the EU have shown a continuous decrease in per capita emissions, their levels remain considerably high, reaching 13.4 t, 7.7 t, and 5.4 t, respectively, in 2024. In comparison, developing countries such as China and India experienced sustained growth in per capita emissions, which climbed to 8.9 t and 2.1 t in the same year. Achieving climate goals requires multifaceted approaches including but not limited to: scaled deployment of CCUS technologies, enhanced energy system efficiency, accelerated renewable energy transition, and innovative negative emission technology systems [52].

2.3. CO_2 emission sources

Fossil fuel consumption dominates global CO_2 emissions (~70 %), primarily through electricity generation, transportation, and industrial activities such as cement and steel manufacturing [49,54]. The energy sector represents the dominant anthropogenic emission source, contributing nearly 33 % of worldwide GHG output. Both mobile (e.g., transportation) and fixed-point sources (e.g., power plants and industrial facilities) substantially add to cumulative emissions. This technological and sectoral heterogeneity demands tailored CCS implementation strategies. Regional emission pattern analysis (Fig. 7) reveals that Eastern China's industrial belt, the US East Coast, developed economies in Western Europe, and the Indian subcontinent form global carbon emission hotspots, with significant national variations in dominant emitting sectors: China focuses on energy-intensive manufacturing, the US emphasizes transportation systems, while the EU is heavily influenced by building heating systems [55].

Research indicates that under current economic and technological

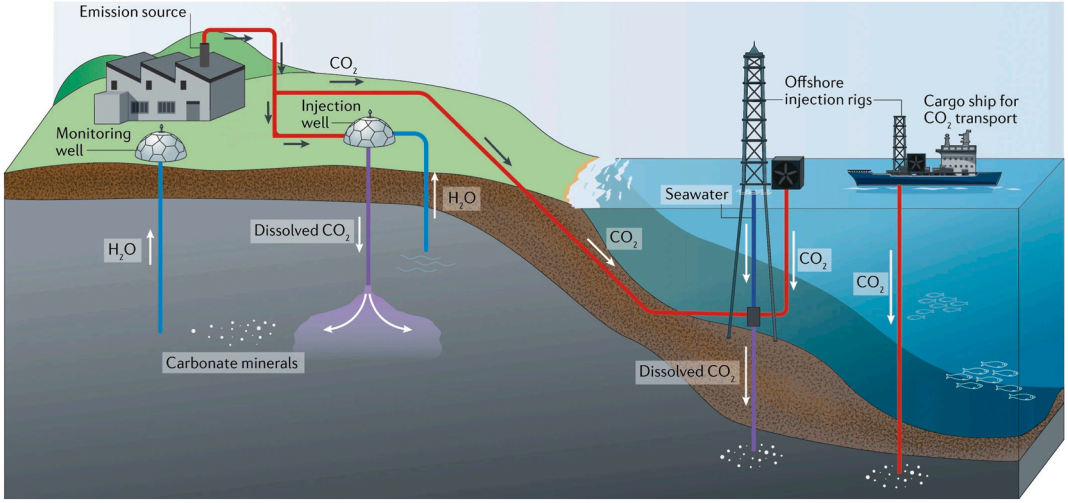


Fig. 2. Schematic diagram of terrestrial and maritime CCUS integrated process (modified after [37]).

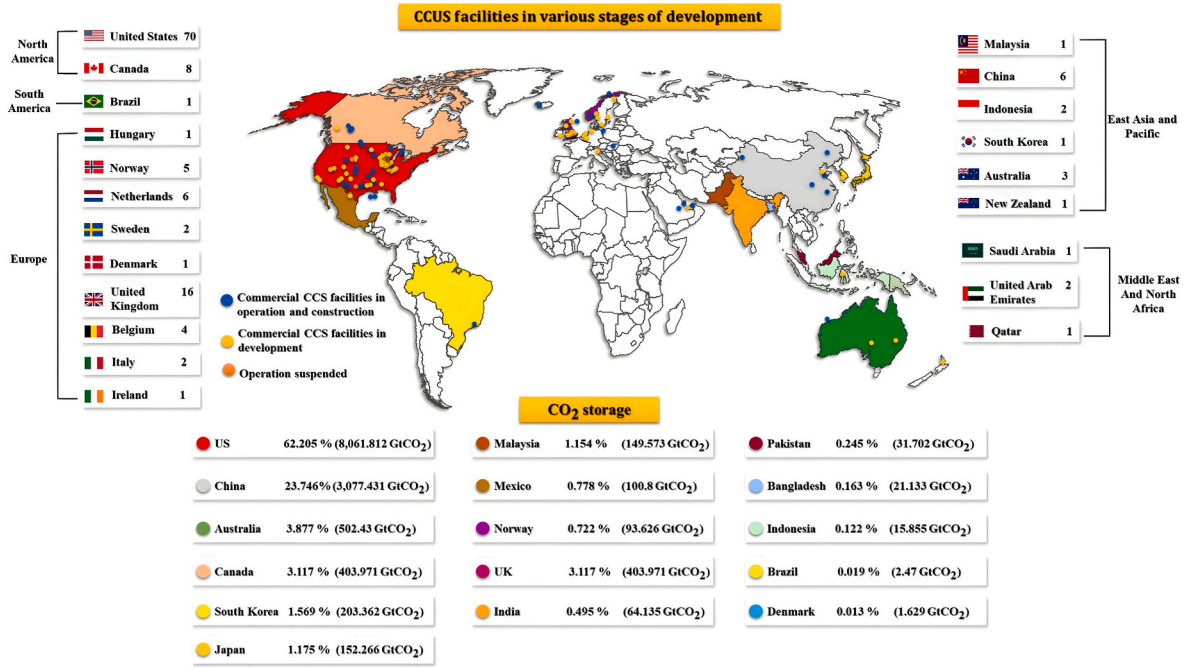


Fig. 3. Global operational CCUS facilities categorized by deployment status (2022) alongside national geological storage potential (2021) [27].

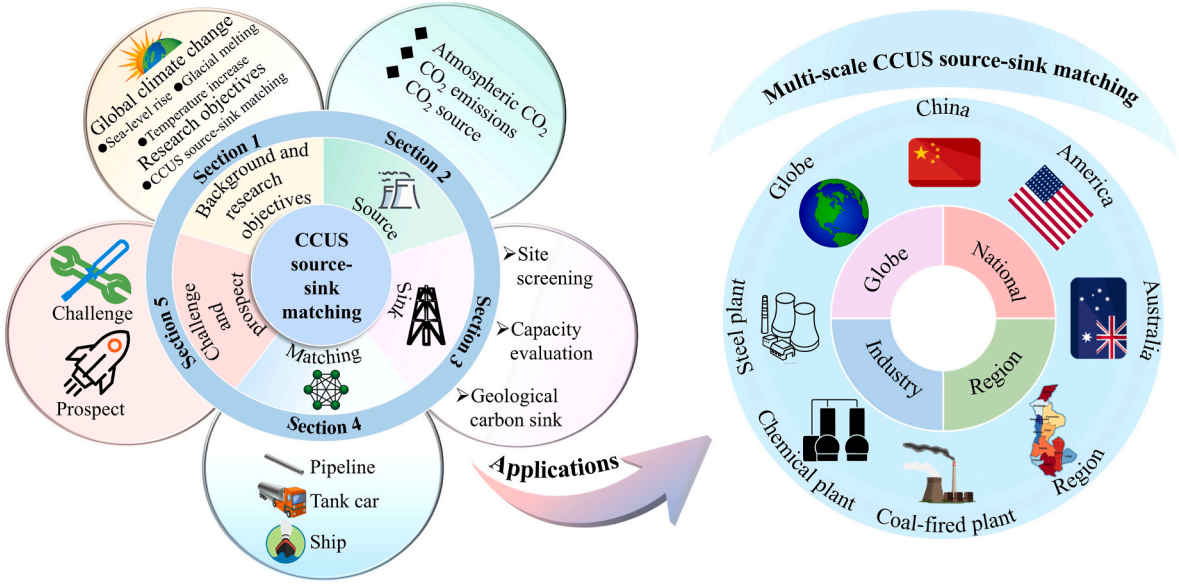


Fig. 4. The research framework of this paper and the specific application fields of CCUS source-sink matching.

conditions, carbon capture for distributed emission sources lacks feasibility for large-scale implementation [56]. Thus, primary fixed-point sources in energy and industrial sectors – including coal/gas-fired power stations, chemical manufacturing plants, petroleum refineries, steel mills, cement production units, and gas processing installations – emerge as principal candidates for accelerating commercial CCS deployment. Their high-density emissions and significant capture scalability drive this strategic prioritization [57–59].

3. Sinks

3.1. Site screening for CO_2 geological storage

3.1.1. Site suitability assessment

Site suitability assessment serves as a prerequisite for CO_2 geological storage (CGS) projects, requiring integrated interdisciplinary approaches from geosciences, engineering mechanics, and environmental sciences to establish a hierarchical evaluation framework [60–62]. This process exhibits phased characteristics: During preliminary assessment, based on storage medium types (e.g., saline aquifers and oil/gas reservoirs), key reservoir parameters including porosity, permeability, and effective thickness are screened through analysis of fluid properties, structural features, and depositional environments, while concurrently

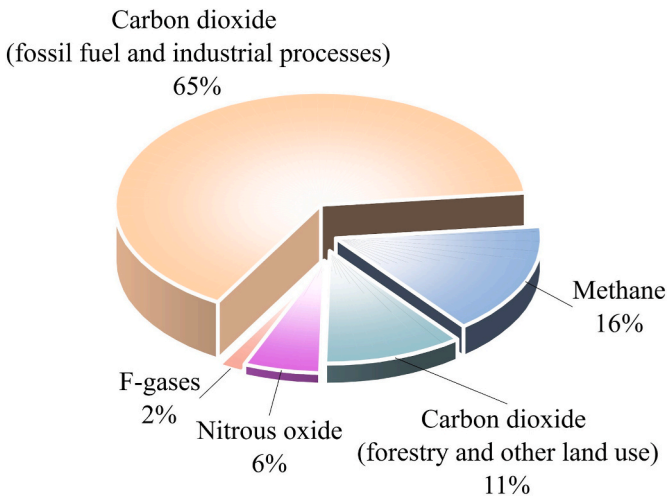


Fig. 5. Types of gases in global GHG emissions.

Table 1
Atmospheric lifetime and GWP values for primary greenhouse gases. CO_2 serves as the reference gas with a standardized GWP of 1. (data from Refs. [43,44]).

GHG		CH_4	N_2O	HFC-23	HFC-134a	SF_6	CCl_4
Life span (years)		12	114	270	14	3200	26
Assessment time of global warming potential	20 years	72	289	12,000	3830	16,300	2700
	100 years	25	298	14,800	1430	22,800	1400
	500 years	7.6	153	12,200	435	32,600	435

evaluating structural trap integrity, caprock breakthrough pressure, and hydrogeological confinement indicators, utilizing multi-parameter weighting models to compute regional-scale storage suitability indices, thereby establishing preliminary screening databases [63]. In intermediate assessment, basin-scale 3D geological modeling techniques are employed, integrating seismic inversion data, core test results, and historical injection cases, focusing on reservoir heterogeneity analysis, pressure transmission boundaries, and mineral carbonation potential, to establish multi-dimensional evaluation matrices incorporating core indicators, enabling quantitative ranking of potential storage units [64]. During final evaluation stages, comprehensive consideration of reservoir geological characteristics (e.g., mineral reactivity), regional infrastructure layout, CO_2 storage policy frameworks, and environmental risk thresholds facilitates the construction of 3D decision-making models encompassing storage efficiency, safety redundancy, and economic feasibility, providing a scientific basis for project implementation [65].

The multi-scale evaluation framework established based on this methodology is illustrated in Fig. 8. The CSLF categorizes geological CO_2 storage suitability assessment into five hierarchical scales: country, basin, regional, local, and site scales [66]. Country-scale assessments focus on contiguous geographic territories within jurisdictional boundaries, typically encompassing multiple sedimentary basins or sub-units, aiming to identify high-potential basins using low-resolution data while evaluating storage types and risk profiles, as exemplified by sedimentary basin studies in China, India, and similar nations [67].

Basin-scale assessments quantify storage potential for specific sedimentary basins, requiring integration of multi-source high-precision data, with the Paraná Basin serving as a representative case [68]. Regional-scale assessments focus on geographically continuous zones within basins exhibiting high source-sink matching or superior storage conditions. such as research on storage capacity in Lithuania's Baltic Basin saline formations and residual oil zones, Malik et al. [69] innovatively developed multi-scenario numerical models, pioneering Monte Carlo uncertainty quantification to calculate theoretical maximum storage capacity: 96–407 Mt in saline formations and 64–267 Mt in oil zones, while revealing critical impact mechanisms of caprock integrity risks and brine migration pathways. Such assessments must avoid active fault zones and mineral rights-dense areas, delineating storage boundaries via 3D geological modeling and pressure field simulations.

Local-scale assessments concentrate on screening candidate reservoirs during pre-engineering phases, systematically verifying reservoir capacity, injection efficiency, and storage performance through numerical simulations coupled with economic analyses [70,71]. Chen et al. [72] utilized Latin hypercube sampling to develop 1000 CO_2 storage-geothermal synergy models based on geological characteristics of northern Oman's foreland basin, combining response surface methodology to quantitatively analyze synergistic effects of reservoir temperature gradients, pore pressure, and other parameters on CO_2 injectivity and thermal extraction efficiency, ultimately establishing multi-objective optimization decision matrices to provide siting foundations for combined storage-geothermal development. Site-scale assessments target specific storage units (e.g., aquifer blocks), requiring construction of multidimensional evaluation systems. Using the Enping 15-1 oil field as an example, Chen et al. [73] established a 9-parameter evaluation system with three indicator layers (e.g., injectivity, storage capacity, and containment security) based on geological data from Field

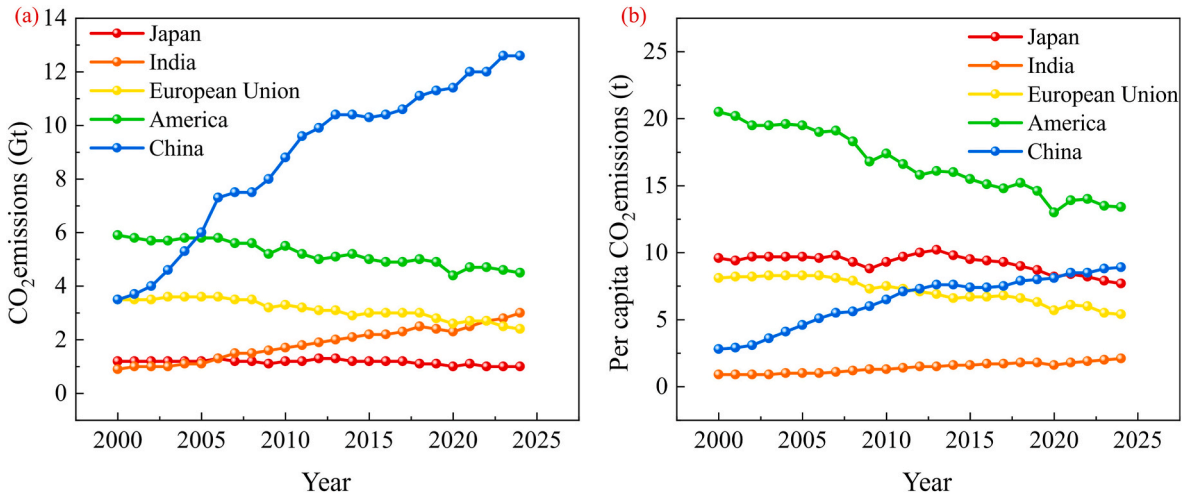


Fig. 6. European Union, China, America, India, and Japan, 2000–2024: (a) total CO_2 emissions; (b) per capita CO_2 emissions (data from Refs. [50,53]).

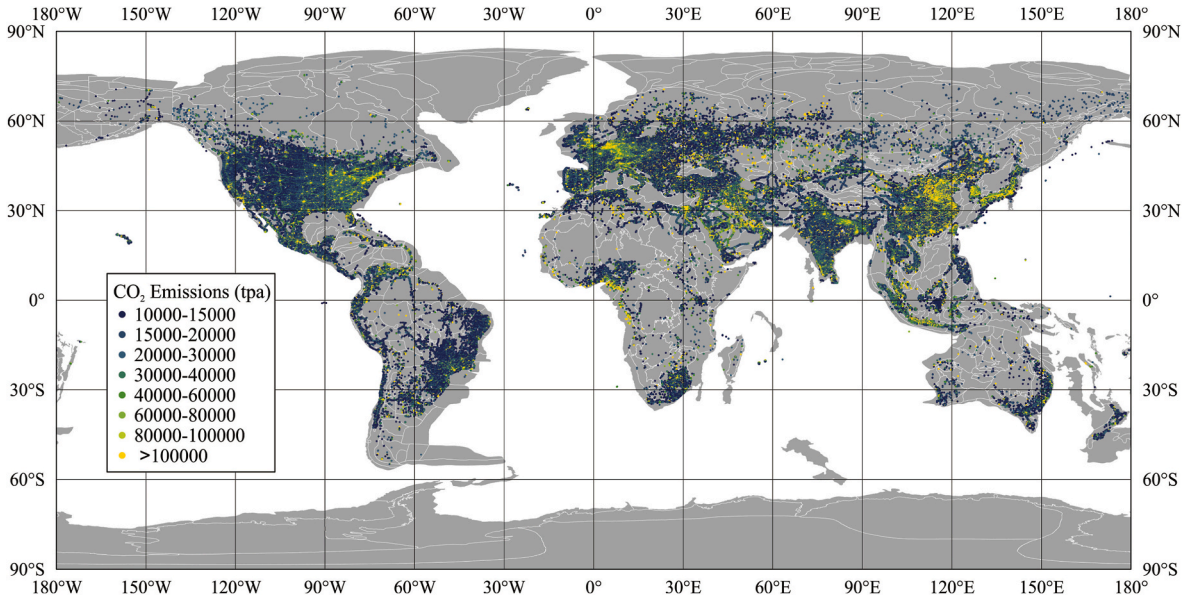


Fig. 7. Distribution of global CO_2 emission sources [55].

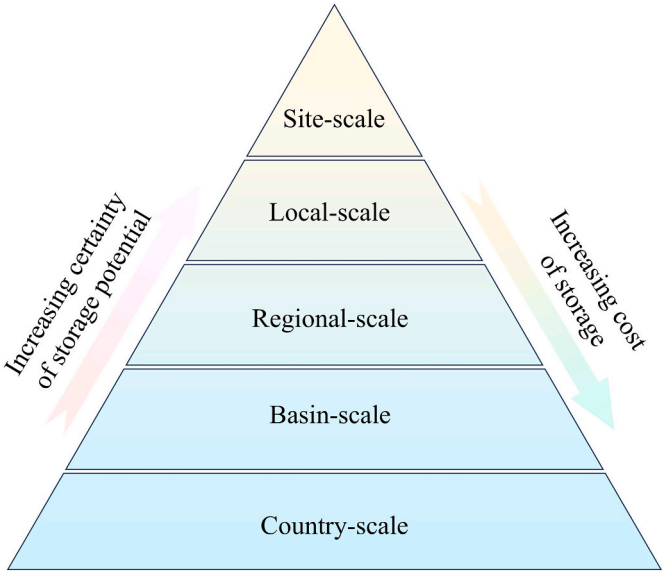


Fig. 8. Suitability evaluation of CO_2 geological storage site.

X's saline formation block, applying analytic hierarchy process (AHP) to identify six dominant factors including porosity and permeability, and creating a nonlinear regression model for storage efficiency, with prediction error (RMSE = 4.32 %) confirming high suitability of target formations.

The assessment system exhibits scale-dependent characteristics: Country and basin-scale assessments are primarily government-led, focusing on macro-scale potential evaluation; local and site-scale assessments are implemented by enterprises, relying on high-resolution 3D geological modeling and multiphase flow coupling simulation technologies; while regional-scale assessments employ public-private data sharing mechanisms to jointly optimize spatial layouts of storage projects and risk regulatory frameworks.

3.1.2. Site screening indicators

Site screening, as a core decision-making component of CCS systems, requires assessment frameworks that deeply integrate geological stability, engineering feasibility, environmental risk controllability, and socioeconomic compatibility [74,75]. Globally, while diverse screening criteria exist across regions due to variations in geological basement characteristics and development stages, they universally adhere to a comprehensive evaluation framework with geological integrity as the foundation, engineering adaptability as the support, risk early-warning mechanisms as safeguards, and socioeconomic acceptance as boundaries [76]. As shown in Table 2, this study systematically synthesizes the influence of various indicators on site screening. The table categorizes key screening parameters into four primary dimensions: geophysical properties, engineering parameters, risk assessment, and socioeconomic factors. Under geophysical properties, critical sub-indicators such as caprock integrity, reservoir configuration, permeability ratios, and pore structure directly determine CO_2 containment security and storage efficiency. Engineering parameters—including injection rate, volume, and wellbore integrity—constrain operational feasibility, while risk and socioeconomic elements such as seismic hazards, leakage potential, public perception, and regulatory policies reflect external uncertainties and social acceptability. This multidimensional framework enables a holistic assessment that balances technical potential with practical deployability, providing a standardized yet adaptable tool for preliminary site selection across diverse geological and regional contexts.

Research demonstrates that intelligent assessment tools based on collaborative optimization of multidimensional parameters can significantly enhance the scientific rigor of preliminary site selection [77–79]. Some scholars have integrated reservoir property analysis with cost-benefit models to establish dynamic decision systems applicable to data-scarce scenarios, providing methodological innovations for large-scale storage potential identification [80–82]. For instance, Ramírez et al. [83] developed a spreadsheet model incorporating storage capacity, cost, and risk management tri-components based on data from 176 Dutch gas fields, oil fields, and aquifers, combining an international expert scoring system to screen CO_2 storage sites, validating site evaluation feasibility under data constraints. This study calculated a total theoretical storage potential of 32 million tons of CO_2. Furthermore, advanced geospatial technology applications reveal dual-directional impact mechanisms of socio-spatial factors on project implementation, emphasizing that non-engineering factors like community engagement and land use conflicts require concurrent quantitative assessment during technical feasibility evaluation. For example, von Rothkirch and Ejderyan [84] analyzed 38 CCS projects through literature review and

Table 2
The influence of various parameters on site screening.

Primary indicator	Secondary indicator	Impact on site screening	Reference
Geophysical properties	Caprock	Caprock petrophysical properties enable long-term CO_2 containment.	[94]
	Anticlinal reservoir configuration	Moderately heterogeneous reservoir structures optimize storage efficiency-safety balance.	[95]
	Sandstone-mudstone interbeds	Dual-seal stratigraphic trap structures mitigate leakage risks.	[96]
	Vertical/ horizontal permeability ratio	High ratios accelerate plume migration but reduce dissolution kinetics; permeability anomalies may trigger leakage.	[97,98]
	Capillary pressure	Rock matrix properties (pore morphology) and thermophysical conditions are critical siting factors.	[99]
	Pore structure	Acicular pores significantly influence elastic parameters versus moldic pores and enhance fracture propensity, improving fluid recovery.	[100]
	Reservoir depth	Depths >800m are required, with increasing depth substantially enhancing storage efficiency.	[101]
	Reservoir thickness	Greater thickness delays leakage, while reduced permeability ratio accelerates CO_2 migration.	[102]
	Geothermal gradient	At >800m depth with 25 °C/km gradient and adequate surface temperature, CO_2 achieves supercritical state.	[103]
	Pressure gradient	Hydrostatic pressure gradient at >800m depth ensures CO_2 reaches the supercritical phase.	[103]
Engineering parameters	Injection rate	Inverse correlation with temperature necessitates optimized injection strategies for site selection.	[104]
	Injection volume	Injection volumes are reservoir-constrained, requiring quantitative performance assessment.	[72]
	Seismic inversion	Identifies storage potential in salt tectonic provinces.	[105]
	Core & log data	Facilitates storage potential assessment in carbonate formations.	[106]
	Wellbore integrity	Prevents casing corrosion and CO_2 leakage.	[107]
Risk assessment and socioeconomic factors	Seismic risk	Integrated geological frameworks and operational parameters determine risk levels to inform site selection.	[108]
	Leakage probability	Permeability anomalies may cause excessive CO_2 leakage, necessitating shallow aquifer monitoring.	[98]
	Fault reactivation	May enhance storage capacity or induce leakage, requiring technical characterization.	[109]
	Public perception	Public awareness management is equally critical as geological assessment for site selection.	[110]
	Regulatory policies	Integrated regulatory considerations reduce carbon abatement costs.	[111]

geographic information system (GIS), finding locational factors benefited 22 projects while hindering 16. This research confirmed GIS applicability for visualizing social factors at potential Swiss storage sites, recommending locational factor assessment during initial siting phases. Industrial cluster analysis demonstrates that optimizing spatial matching between emission sources and storage sites can significantly enhance carbon transport network economics. However balancing geological storage capacity with abatement demands is essential [85,86]. For instance, Fernandez-Canteli alvarez et al. [87] employed industrial clustering methodology integrating environmental assessment, socio-economic factors, and techno-economic analysis to study CCUS implementation in Spain's Ebro basin. The study found that a baseline scenario comprising 15 major emitters could form a complete transport network but requires holistic consideration of storage capacity constraints and abatement cost-effectiveness. Callas et al. [88] innovatively developed a three-phase quantitative assessment system through capacity-injectivity optimization, risk minimization, and economic constraint analysis, screening 10 optimal clusters comprising 31 oil fields from 1317 depleted reservoirs in the Gulf of Mexico.

Current technological advancements show multidisciplinary convergence, where integrated quantitative risk assessment systems and geo-economic collaborative models are reshaping traditional siting paradigms [89,90]. Comprehensive decision frameworks formed by coupling surface deformation monitoring, subsurface fluid migration simulation, and carbon market mechanism analysis, not only expand the spatiotemporal dimensions of site screening but also achieve organic integration of environmental and economic benefits, providing new theoretical foundations and practical pathways for carbon storage project deployment under global carbon neutrality goals [91–93].

3.2. *Capacity assessment for CO_2 geological storage*

As a core technological pathway for mitigating rising atmospheric CO_2 concentrations, geological storage capacity assessment must comprehensively consider multidimensional constraints, including lithospheric physicochemical properties, structural trap integrity, and long-term containment security [112–114]. Globally, unminable coal seams, depleted oil and gas reservoirs, and deep saline formations act as primary CO_2 storage media. All rely on the four core CO_2 sequestration mechanisms—mineral trapping, dissolution trapping, residual trapping, and structural trapping—but differ markedly in each mechanism's dominance: deep saline formations utilize all four synergistically, with structural trapping ensuring initial confinement, residual trapping immobilizing pore CO_2, dissolution trapping via CO_2 dissolution in brine, and mineral trapping enabling long-term carbonate mineral fixation [115]; depleted oil and gas reservoirs primarily rely on structural trapping (leveraging their well-verified geological traps) and residual trapping, supplemented by dissolution trapping, while mineral trapping serves as a secondary long-term mechanism [116]; unminable coal seams require structural trapping as a prerequisite, with limited residual and dissolution trapping, and their dominant process is CO_2 adsorption (physical/chemical adsorption on organic matrices), while mineral trapping is comparatively weak [117]. These storage processes fundamentally depend on these four mechanisms to achieve permanent CO_2

sequestration [118–120]. As illustrated in Fig. 9, structural trapping prevents CO_2 migration using physical confinement beneath low-permeability sealing layers (e.g., shale or mudstone) [121]. Within porous media, residual trapping immobilizes CO_2 through capillary entrapment, forming isolated droplets [122]. Solubility trapping diminishes leakage potential when CO_2 dissolves into the formation water's aqueous phase [123], whereas mineral trapping ensures permanent sequestration via geochemical conversion to stable carbonates [124].

In response to differential storage mechanisms across geological media and spatiotemporal scales, the academic community has developed targeted capacity assessment frameworks. Sang et al. [125] established a three-tier capacity index system (effective storage capacity, theoretical storage capacity, and maximum storage capacity) for deep coal seam storage systems. The CO_2BLOCK tool, developed by De Simone and Krevor [126], enables refined assessment of structural reservoir storage potential under varied well patterns through dynamic pressure constraint analysis. Leveraging China's 13th five-year plan hydrocarbon resource data, Li et al. [127] implemented an enhanced four-dimensional evaluation model (P-V-T equation of state integrated with CSLF standards), systematically evaluating layered storage potential across 29 petroliferous basins and establishing priority development sequences for continental basins (e.g., Songliao and Bohai Bay) and marine basins (e.g., Bohai and Pearl River Mouth). Kamolov et al. [8] employed resource assessment and spatial analysis to evaluate CO_2 storage-utilization potential in Uzbekistan's power and cement sectors, finding annual storage capacity reaches 1.171 billion tons of CO_2—far exceeding emissions—yet chemical utilization prospects are limited, with geographical mismatches between emission sources and storage sites.

As shown in Table 3, capacity calculation methods vary substantially across storage media, primarily including material balance, volumetric, areal, dissolution mechanism, and solubility approaches. These methods all yield theoretical storage estimates, but their applicability and reliability vary with assessment scale, storage type, and data availability: For oil/gas reservoirs and deep coal seam assessments, material balance methods prove more reliable than volumetric approaches and are applicable at finer scales; whereas for saline formations, volumetric methods demonstrate higher accuracy and broader applicability than dissolution mechanism/solubility methods. Table 3 systematically summarizes these key methodologies, along with their governing equations, critical parameters, and primary proponents such as CSLF, US-DOE, and other research institutions. Each method is tailored to specific reservoir types and trapping mechanisms: for instance, the material balance method incorporates factors like hydrocarbon recovery and cumulative water production for depleted reservoirs, while volumetric approaches apply storage efficiency factors tailored to saline aquifers. The selection of an appropriate method depends not only on the geological context but also on data quality and project objectives, underscoring the necessity of a multi-method assessment strategy to reduce uncertainty and enhance the credibility of capacity estimates at

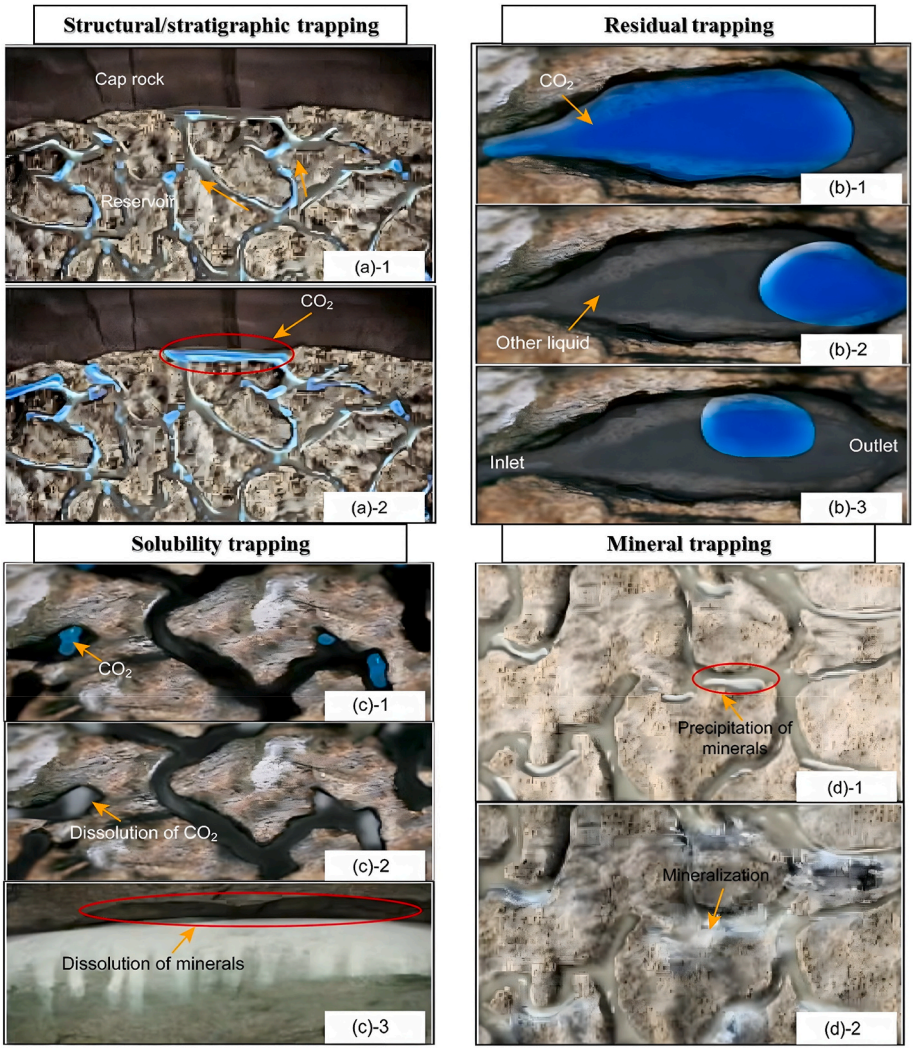


Fig. 9. Four major trapping mechanisms in CGS. (a) structural; (b) residual; (c) solubility; and (d) mineral trapping [27].

Table 3
Primary methodologies for quantifying CO_2 storage capacity in confined geological formations.

Method type	Proposer	Storage type	Formula	Parameter description	References
Material balance method	CSLF	Oil and gas reservoirs	$M_{CO2,oil} = \rho_{CO2}[R_f Ah\varphi(1 - S_w) - V_{iw} + V_{pw}]$	$M_{CO2,oil}$: Geological CO_2 storage capacity within oil reservoirs (kg); ρ_{CO2}: CO_2 density at reservoir conditions (kg/m^3); R_f: Hydrocarbon recovery factor (%); A: Areal extent of reservoir (m^2); h: Effective thickness (m); φ: Mean porosity (%); S_w: Average reservoir water saturation (%); V_{iw}/V_{pw}: Cumulative water injection/production volume (m^3)	[128]
	CSLF	Unminable coal seams	$M_{CO2,coal} = \rho_{CO2} Ahn_c G_c(1 - f_a - f_m)$	$M_{CO2,coal}$: CGS capacity in unminable coal seams (kg); n_c: Volumetric gas content of coal (kg/m^3); G_c: Gas content of coal seam (m^3/kg); f_a/f_m: Ash and moisture content of coal (%)	[128]
	CSLF	Saline reservoir	Structural trapping: $M_{CO2,g} = \rho_{CO2} Ah\varphi(1 - S_{wirr})$ Residual trapping: $M_{CO2,r} = \rho_{CO2r} S_{CO2r}\varphi\Delta V$ Solubility trapping: $M_{CO2,d} = (\rho_s X_{sCO2} - \rho_i X_{iCO2})Ah\varphi$	$M_{CO2,g}$: Structural trapping capacity (kg); S_{wirr}: Irreducible water saturation; $M_{CO2,r}$: Theoretical storage capacity for residual trapping (kg); ΔV: Rock volume previously CO_2−saturated then water−flooded; S_{CO2r}: CO_2 saturation after brine re−influx; $M_{CO2,d}$: Theoretical storage capacity for solubility trapping (kg); ρ_s/ρ_i: Average densities at saturated/initial states; X_{sCO2}/X_{iCO2}: CO_2 mass fractions in formation water.	[128]
Volumetric method	US-DOE	Oil and gas reservoirs	$M_{CO2,oil} = \rho_{CO2} Ah\varphi(1 - S_w)BE$	B: Formation volume factor (ratio of hydrocarbon volume at reservoir conditions to surface volumes, dimensionless); E: Storage efficiency factor	[129]
	US-DOE	Unminable coal seams	$M_{CO2,coal} = \rho_{CO2m} AhC_{smax}E$	ρ_{CO2m}: CO_2 density at standard conditions (kg/m^3); C_{smax}: Maximum CO_2 adsorption capacity per unit coal volume at standard conditions (0 °C, 101325 Pa)	[129]
	US-DOE	Saline reservoir	$M_{CO2,saline} = \rho_{CO2} Ah\varphi E$	$M_{CO2,saline}$: CGS in saline reservoirs (kg)	[129]
Areal method	EC	Saline reservoir	$M_{CO2,saline} = ACF \times A \times SF$	ACF: Aquifer coverage factor; SF: Storage coefficient (kg/m^2)	[130]
Dissolution mechanism method	BACHU & ADAMS	Saline reservoir	$M_{CO2,saline} = (\rho_s X_s^{CO2} - \rho_i X_i^{CO2})Ah\varphi C$	X_s^{CO2}/X_i^{CO2}: CO_2 mass fractions in formation water under saturated/initial states; C: diffusion-dissolution factors	[131]
Dissolution mechanism method	Shen et al.	Oil and gas reservoirs	$M_{CO2,oil} = \rho_{CO2}[Ah\varphi(1 - S_{wi}) - V_{iw} + V_{pw} + C_{ws}(Ah\varphi S_{wi} + V_{iw} - V_{pw}) + C_{os}(1 - R_f)Ah\varphi(1 - S_{wi})]$	S_{wi}: Irreducible water saturation; C_{ws}: CO_2 solubility in water; C_{os}: CO_2 solubility in oil	[132]
Solubility-based method	Li et al.	Saline reservoir	$M_{CO2,saline} = aA\rho_w MnhR\varphi$	a: Areal fraction of CO_2-suitable saline formations (default = 0.01); n: Effective thickness ratio for aquifers (default = 0.1); R: CO_2 solubility coefficient in brine; ρ_w: Brine density under storage conditions (kg/m^3); M: CO_2 molar mass (g/mol)	[133]

regional to site-specific scales.

According to China's CCUS 2021 annual report estimates, global onshore geological CO_2 storage capacity ranges from 6–42 trillion tons, while offshore potential reaches 2–13 trillion tons, with deep saline formations contributing 98 % of global storage capacity [21]. Fig. 10 displays geological CCUS storage potential versus CO_2 emissions for major countries and regions worldwide (Table 4). As Asia's core storage region, China's total potential approximates 1.21–4.13 trillion tons. Specifically: Four major basins including Songliao, Bohai Bay, Ordos, and Junggar Basins could achieve 5.1 billion tons of storage via CO_2-EOR in oil fields; Ordos, Sichuan Bohai Bay and Tarim Basins possess gas reservoir capacities of 15.3 and 9.0 billion tons respectively using depleted reservoirs with CO_2-EGR technology; China's saline formation storage totals about 2.42 trillion tons, with three core zones—Songliao (694.5 billion tons), Tarim (552.8 billion tons), and Bohai Bay (490.6 billion tons) basins—comprising 50 % collectively. Other Asian regions exhibit total potential of 0.49–0.55 trillion tons. Herein, Japan's 140-billion-ton capacity primarily resides in offshore sedimentary basins like Tokyo Bay and Osaka Bay; Korea's Jeju basin oil reservoirs contribute 23.5 billion tons while North Pyongyang basin aquifers account for 0.9 Gt; the total storage potential of Indonesia, Thailand, the Philippines, and Vietnam is approximately 54 billion tons.

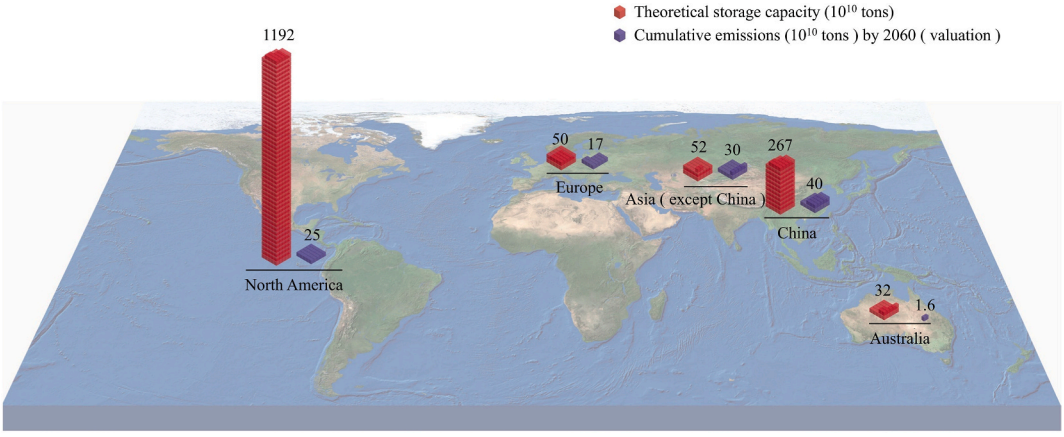


Fig. 10. Global assessment of CCUS geological storage potential versus CO_2 emissions across key regions (storage capacity values represent median interval estimates) (modified after [21]).

Table 4
CCUS geological storage potential and CO_2 emissions in major countries and regions of the world [21].

Country/ region	Theoretical storage capacity (10^{10} tons)	Estimated cumulative CO_2 emissions by 2060 (10^{10} tons)
North America	230~2153	25
Europe	50	17
Asia (except China)	49~55	30
China	121~413	40
Australia	22~41	1.6

North America's total storage potential reaches 2.3–21.53 trillion tons, displaying distinct tri-national distribution: The United States Geological Survey (USGS) evaluates 36 U S. basins averaging 3 trillion tons, while the Global CCS Institute (GCCSI) estimates 2–21 trillion tons total potential; Canada's petroliferous basins hold 16 billion tons potential, with saline formations at 0.028–0.296 trillion tons; Mexico's saline formations exceed 0.1 trillion tons potential. Per the North American Carbon Sequestration Atlas (NACSA) data, U.S. and Canadian saline formation potentials reach 16.1–202 and 0.28–2.96 trillion tons respectively, constituting regional storage anchors. Europe's total potential approximates 0.5 trillion tons with pronounced spatial distribution: North Sea, western and eastern European petroliferous basins possess 30 billion tons storage potential; The west-east Europe tectonic belt saline formations hold 0.325 trillion tons potential. Additionally, Anthonsen et al. [134] assessed 25 nations' hydrocarbon reservoirs, saline formations, and coal seams totaling 0.126 trillion tons potential. The European Union (EU) GeoCapacity project confirmed deep saline formations constitute 65 % of total storage, highlighting their central role in European carbon sequestration strategy. Australia's geological potential ranges 0.22–0.41 trillion tons, with 65 CO_2-suitable sedimentary structures predominantly distributed in coastal and central regions.

Current assessments indicate global geological CO_2 storage capacity reaches hundred-billion-ton scale, significantly exceeding cumulative global emissions, providing theoretical feasibility for large-scale CCUS deployment. Global deep coal seams and oil/gas reservoirs each possess hundred-billion-ton scale CO_2 storage capacity. Though substantially smaller than saline formations, CO_2-enhanced gas recovery (CO_2-EGR) and CO_2-enhanced oil recovery (CO_2-EOR) storage demonstrate greater economic viability and safety, particularly as CO_2-EOR dominates current projects, enhancing the practical significance of hydrocarbon and unminable coal seams storage. Given saline formations' immense potential, they will receive increased focus and are poised to become the most promising geological CO_2 storage type.

3.3. CO_2 geological sinks

CO_2 geological sinks refer to technological systems that actively inject atmospheric CO_2 into subsurface formations through natural processes or engineered interventions [135,136]. Core pathways include mineral carbonation reactions [137], karst carbon sink effects [138], and oceanic dissolution absorption [139], along with deep geological storage—the most scalable mitigation approach—permanently sequestering CO_2 in supercritical state (>31.1 °C, >7.8 MPa) within saline formations, depleted reservoirs, or deep strata beyond 800 m depth [140,141]. As shown in Fig. 11, current global CGS and utilization technologies exhibit tiered development: CO_2 storage coupled with in-situ uranium leaching (CO_2-ILU) has achieved commercial application, where China, Russia, USA, and Kazakhstan inject CO_2 into sandstone uranium deposits, dissolving uranium oxides (UO_2^{2+}) in acidic conditions, significantly enhancing uranium recovery while concurrently storing CO_2 [142]. CO_2-EOR technology has entered commercial implementation phase, operating by injecting supercritical CO_2 into reservoirs to reduce crude viscosity and swell oil phase volume [143,144]. When injection pressure surpasses minimum miscibility pressure (MMP), CO_2 and crude oil form miscible displacement fronts, substantially improving oil recovery (typically by 8–15 %) while enabling structural and dissolution trapping [145–147]. As depicted in Fig. 12, the integrated CO_2-EOR process incorporates both external and internal carbon cycles. The external cycle utilizes CO_2 captured from industrial sources, which is transported and injected into oil reservoirs to enhance recovery, while the internal cycle involves reprocessing and reinjecting CO_2 separated from produced oil and gas. This closed-loop system reduces reliance on external CO_2 and improves storage efficiency, with a portion of CO_2 permanently trapped. Zhang et al. [148] supported its feasibility through a techno-economic model based on six oilfield blocks, showing a break-even oil price of $29.4/barrel with carbon tax and $25.6/barrel without, alongside a 54.7 % emission reduction, confirming both economic and environmental benefits.

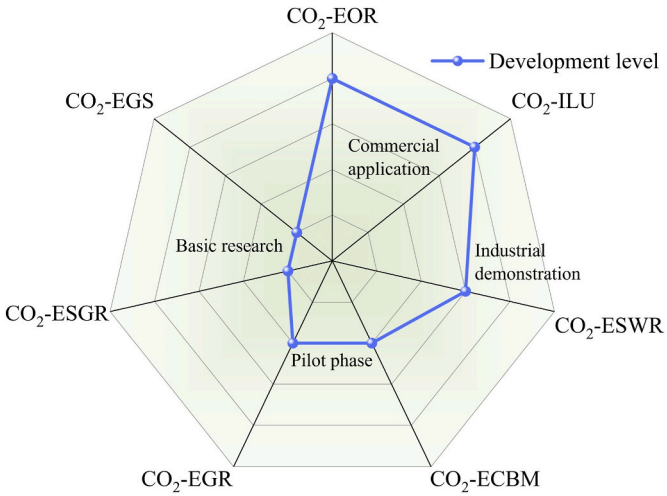


Fig. 11. The development level of global CO_2 geological storage and utilization by 2022 (data from Ref. [21]).

CO_2-enhanced saline water recovery (CO_2-ESWR) is at the industrial demonstration stage, utilizing CO_2 to displace brine in deep saline formations while producing industrial water via extraction wells (e.g., Shenhua project in Ordos Basin, China), achieving permanent storage through capillary trapping (residual trapping) and mineral carbonation (forming siderite/calcite) [149]. This technology faces plume migration risks from reservoir heterogeneity, requiring management through real-time pressure monitoring at injection wells and caprock integrity assessment [150–152]. By contrast, CO_2-EGR and CO_2-enhanced coalbed methane (CO_2-ECBM) remain at pilot testing stage: The former relies on CO_2's strong adsorption in coal micropores to displace methane, but Qinshui basin tests in China revealed permeability reduction from coal swelling [153,154]; The latter maintains reservoir pressure and displaces residual gas via CO_2 injection into depleted reservoirs, exemplified by the Dutch K12-B project verifying high-purity CO_2 injection feasibility, yet faces challenges of early CO_2 breakthrough and low storage efficiency [155,156]. Meanwhile, CO_2-enhanced shale gas recovery (CO_2-ESGR) and CO_2-enhanced geothermal systems (CO_2-EGS) remain in a fundamental research phase, with CO_2-EGS requiring solutions for controlling CO_2-water-rock reactions in fractured hot rock masses [157,158], and CO_2-ESGR needing optimization of supercritical CO_2 flow mechanisms in nanoscale shale pores [159].

However, large-scale geological carbon sink deployment currently requires solving the "source-sink mismatch" challenge [160]. Global spatial imbalance between emission sources and storage sites presents a key challenge: In China, most saline storage occurs in western regions (e.g., Songliao and Tarim basins) while primary emissions concentrate in eastern coastal areas [161]; Europe and America achieve cross-border matching via North Sea carbon hubs (e.g., German industries linking

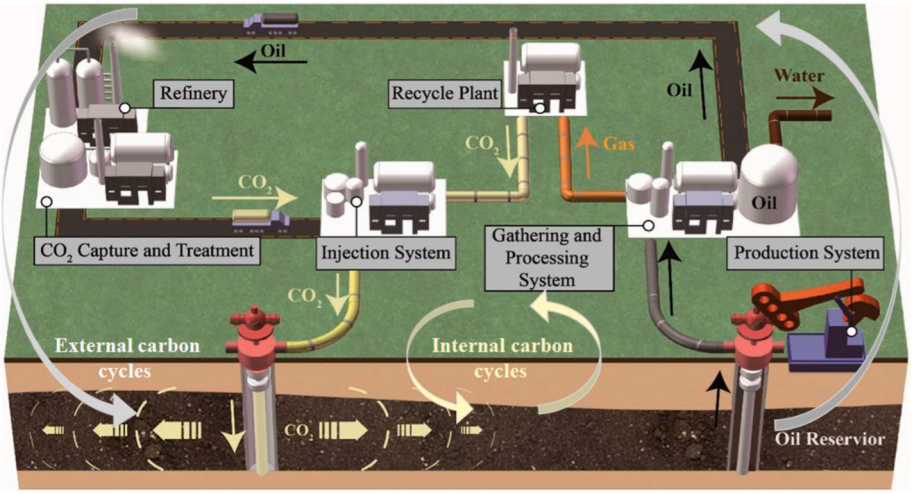


Fig. 12. The whole process application of CO_2-EOR technology [148].

to Norwegian storage), whereas southeast Asia lacks comparable collaborative networks [162]. Current research must integrate pipeline network optimization, geological risk assessment, and policy coordination to develop cross-scale "source-sink-pipeline" dynamic matching models.

4. Source-sink matching

4.1. Transportation modes

As a critical component of the CCUS chain, CO_2 transport mode selection directly impacts project economics and risk profiles [163,164]. Primary transport modes include pipelines, shipping vessels, and tanker trucks (Table 5). Pipeline transport dominates large-scale, long-distance scenarios [165–167]. Analysis of four Germany-to-Norway cross-sea transport scenarios reveals transport cost share decreases from 22 % to 10 % as annual capture volume increases; shipping proves more economical at small scales while pipelines excel at large scales, though scale economies must be balanced against capital lock-in effects [168]. This cost differential stems from pipeline transportation's economies of scale. China's multi-sector shared pipeline model indicates 19,000 km of trunk pipelines will be needed by 2050 to support 174 million tons/year carbon capture, with 50 % reused from existing oil/gas corridors [169]. However, pipeline transport must overcome supercritical phase control challenges. Jiao et al. [170] demonstrated that implementing dense-phase transport in downstream supercritical pipelines enhances CO_2 storage by 3.4 % through pressure/temperature control coefficients, confirming that transport constraints significantly impact source-sink matching optimization. Furthermore, as Fig. 13 shows, offshore integrated CCS value chains require emphasis on pipeline design, network optimization, phase-change control, and corrosion prevention [171].

Maritime shipping offers unique advantages for small-to-medium scale projects. Norway's Brevik cement plant achieved minimum transport costs of €33.8/ton using a dual-stage buffered storage strategy (18 % capacity redundancy) when addressing North Sea weather volatility, representing 5 % cost reduction versus non-buffered alternatives [172]. Innovative international shipping models are emerging: Datta et al. [16] proposed a dual-use model utilizing liquefied natural gas (LNG) carriers for backhaul CO_2 transport, developing a CCUS economic model to examine the feasibility of using Japanese/Korean CO_2 for US offshore EOR, and discussing policy frameworks to facilitate international CO_2 trade. However, vessel transport faces scalability limitations, Bennaes et al. [173] conducted mixed-integer programming models to optimize vessel logistics based on Europe-to-Norway carbon transport demand, showing that under 2050 high-demand scenarios, per-ton transport costs could drop to €8, requiring about 100 vessels for thousands of annual voyages.

Tanker truck transport serves as a supplementary solution for small-scale scenarios, primarily for temporary deployment or pilot projects. Its core challenges lie in pressure vessel safety and long-term transport economics [174]. Although composite materials can reduce vessel rupture risks, per-unit transport costs increase nonlinearly with distance, making it difficult to adapt to long-distance, continuous transport demands difficult [175]. Current research indicates its role in large-scale CCUS deployment is confined to such specialized contexts.

Current CCUS transport systems still confront multidimensional technical bottlenecks: First, risk assessment suffers from weak data foundations, with CO_2 pipeline corrosion models exhibiting significant uncertainty due to limited historical incident records, particularly regarding impurity concentration and phase-change impacts on pipeline integrity requiring deeper investigation; Second, multimodal coordination mechanisms are absent, with no established cost-risk tradeoff models for integrated pipeline-vessel-tanker operations, hindering global optimization for cross-regional source-sink matching; Third, geological adaptability is insufficient, as pipeline networks for complex reservoirs (e.g., saline formations and abandoned coal fields) are vulnerable to subsurface heterogeneity, while stacked storage strategies enhance flexibility but incur additional costs [176]. Moreover, policy frameworks lag, with international shipping relying on cross-border carbon tariffs and storage liability allocation rules, yet current regimes inadequately support innovative models like dual-purpose vessels.

Future development pathways should focus on three key directions: Technologically, develop intelligent pipeline networks integrating real-time geological variability data [177], pressure plume simulations, and sink clustering algorithms [178], to achieve dynamic route optimization and risk prediction; For cross-modal integration, establish full-chain digital twins covering "capture-transport-storage", quantifying weather/geological uncertainties' impact on multimodal costs via two-stage stochastic programming; Institutionally, urgently construct international carbon transport policies, clarify cross-border storage liability mechanisms, promote repurposing standards for existing oil/gas infrastructure [160], and explore financial instruments to hedge weather-induced storage cost fluctuations. Long-term, advanced interdisciplinary integration of materials science (e.g., corrosion-resistant alloys and composite vessels) and supercritical control technologies, overcoming phase-state control bottlenecks, ultimately forming a trinity synergistic evolution framework integrating technology, economics, and governance.

Table 5
Comparative analysis of CO_2 transport modes for CCUS: pipelines, shipping vessels, and tanker trucks.

Dimension	Pipelines	Shipping vessels	Tanker trucks
Primary use case	Large-scale, fixed sources/sinks, continuous flow, long-term projects	Medium-scale, intercontinental/ island transport, intermittent supply, hub-and-spoke models	Small-scale, temporary/ emergency supply, last-mile delivery, pilot projects
Technology maturity	Commercial	Commercial	Commercial
Key risk factors	Corrosion, geohazards (e.g., seismic and landslides), third-party interference, and permeation loss	Weather delays, sloshing/boil-off management, port accessibility, piracy/security, bunker fuel volatility	Road accidents, pressure vessel fatigue/failure, rollover risk, loading/ unloading incidents
Cost structure	Very high CAPEX (material, construction), low OPEX/km/ton, strong economies of scale	Moderate-high CAPEX (vessel), high variable OPEX (e.g., fuel and port fees), economies of scale on vessel size	Low CAPEX (per unit), very high OPEX/ton-km, limited economies of scale
Technical optimization focus	Supercritical/ dense-phase operation, corrosion-resistant alloys, smart pigging, topology optimization for geological constraints, hybrid (H_2/CO_2) pipelines	Dedicated CO_2 carriers (varying pressures), LNG dual-fuel propulsion, onboard reliquefication, optimized cargo handling systems, intermediate storage buffers	Advanced composite tanks (Type IV/V), weight reduction, cryogenic systems for liquid CO_2, automated loading/ unloading
Environmental impact	Significant land disruption during construction, low operational emissions (if powered renewably), potential for offshore seabed impact	Higher operational GHG emissions (bunker fuel), ballast water management, port area emissions, and underwater noise	Highest per-ton operational emissions (diesel fuel), local air pollution (e.g., particulates and NOx), traffic congestion
Geographical constraints	Requires stable geology, faces challenges with mountains, water bodies, urban areas, and permafrost	Requires deep-water ports, constrained by navigational channels, ice conditions, and geopolitical chokepoints	Limited to land-based routes with suitable road infrastructure, max. practical range <1000 km
Infrastructure requirements	Dedicated, fixed infrastructure (e. g., compressor stations and valves)	Port facilities (e.g., loading/unloading arms and storage tanks), bunkering infrastructure	Road network, simple loading bays, no fixed route
Operational flexibility	Low (fixed route/ source/sink), complex/costly to reroute	Medium (e.g., flexible routing and multiple potential ports), schedule-dependent	Very High (on-demand, point-to-point, easy rerouting)
Regulatory landscape	Complex long-term permits, cross-jurisdictional harmonization needed	International maritime regulations, port state controls, and emissions trading schemes	National/regional road transport regulations, weight/size limits, driver certifications

4.2. Source-sink matching methods

Source-sink matching represents the core process in CCUS system design, optimizing spatial, scale, and dynamic connectivity between CO_2 emitters and storage reservoirs (Fig. 14). International mainstream methods primarily employ mathematical programming, GIS spatial analysis, and intelligent optimization algorithms (Table 6), evolving significantly from single-stage static matching to multi-phase dynamic optimization. Within mathematical programming, mixed-integer linear programming (MILP) serves as a foundational tool by concurrently handling discrete decisions (e.g., pipeline construction) and continuous variables (e.g., flow allocation) [179]. For instance, the SimCCS model series developed by Los Alamos National Laboratory pre-generates candidate networks permitting pipeline merging and branching, substantially reducing computational complexity for multi-source/multi-sink matching; Its upgraded SimCCS TIME version incorporates temporal dimensions, simultaneously optimizing capture locations, volumes, and scheduling to enable dynamic planning for an integrated CCUS cluster [180]. The EU's InfraCCS model simplifies problems into mixed-integer formulations, achieving cross-sectoral co-optimization through integration with the MARKAL energy systems platform [181]. For nonlinear challenges (e.g., compressor costs and geological constraints), two-stage linearization methods transform complex cost functions into solvable MILP formulations. Al-Mohannadi et al. [182] applied MILP decomposition to linearize pipeline/compressor costs, constructing multi-period optimization models for carbon network design, rapidly solving renewable-integrated CCUS systems to deliver cost-optimized decarbonization policy solutions. Multi-objective mixed-integer nonlinear programming (MINLP) combined with stochastic programming effectively manages uncertainties, as Qiu et al. [183] demonstrated using Xinjiang CO_2-EOR networks, solving multi-objective MINLP through piecewise linearization and augmented ε-constraint methods, validating its advantages in optimizing pumping station costs, injection rates, and supply-demand deviations. Recent innovations focus on mechanism integration, such as Tsinghua University's ChinaCCS DSS model. Its initial version generated transport-storage cost curves using greedy algorithm bubble sorting; Version 2.0 upgraded to an MILP framework supporting complex scenarios with multiple sources connected to multiple sinks and shared pipelines [184].

Spatial analytics and intelligent algorithms provide critical complementary support to mathematical programming [185]. Spatial analysis enables rapid macro-screening via GIS, exemplified by the Chinese Academy of Sciences' ITEAM-CCUS approach, which generates levelized-cost-optimal source-sink pair sequences through national storage potential datasets and cost matrices [186]. Moreover, multi-dimensional decision integration methods continue emerging, for instance, Zhou et al. [187] developed GIS-DEMATEL-ISM framework identifying technological gaps and policy deficiencies as fundamental barriers to China's CCUS development. As a core spatial tool, least-cost path analysis (LCPA) computes terrain-adjusted optimal routes using Dijkstra's algorithm, enabling decentralized deployment in trunk-pipeline-free regions: Chen et al. [188] applied this method to plan minimum-cost pipeline networks connecting 88 Hebei emission sources to 25 storage sinks. Intelligent algorithms demonstrate robust modeling and solving capabilities for source-sink matching optimization. For instance, fuzzy decision theory significantly enhances storage site evaluation reliability by handling uncertainties in siting assessments [189]; Graph network optimization coordinates environmental constraints and engineering conditions to achieve globally optimal cross-regional pipeline planning [111]; Genetic algorithms effectively resolve multivariate optimization of pipeline parameters and compressor station layouts under complex terrain through parallel search mechanisms [190]. These approaches overcome limitations of conventional mathematical programming in high-dimensional nonlinear systems and provide intelligent decision support balancing technical feasibility and economic optimality for system-level source--sink matching. Additionally, there are also some non-mainstream methods, such as pinch analysis, the system optimization method for minimizing transportation cost, and the distributed connection method [191–193].

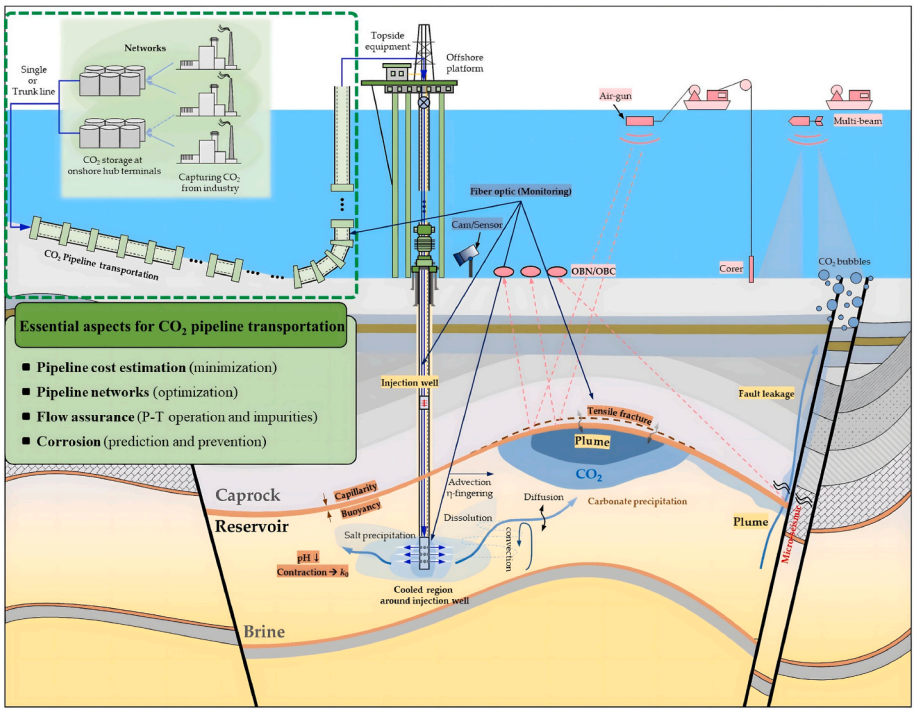


Fig. 13. The integrated CCS value chain diagram of the maritime case (modified after [171]).

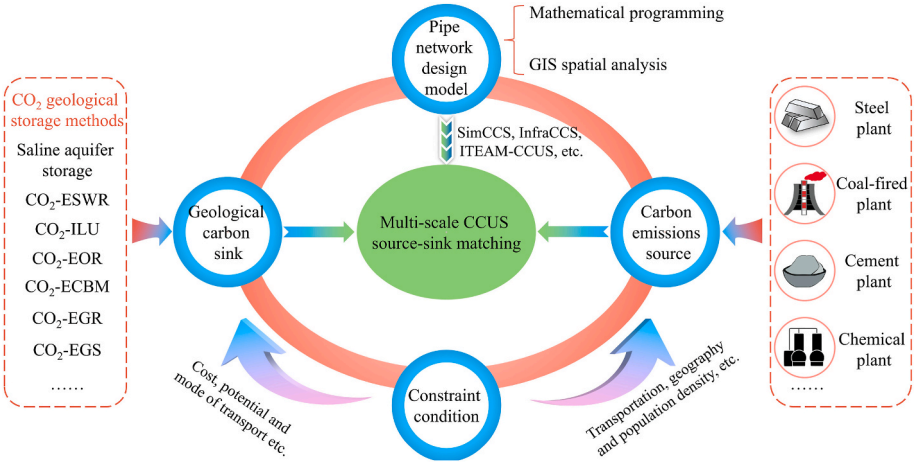


Fig. 14. CCUS source-sink matching conceptual model.

Current methodologies exhibit a new "temporal-spatial-constraint" tri-integration paradigm: Temporally, models like SimCCS TIME and InfraCCS advance static planning toward multi-period dynamic optimization; Spatially, deep GIS-mathematical programming integration (e.g., high-resolution grid-driven MILP) enhances pipeline engineering precision; In constraint dimension, stochastic programming handles geological variability, neutrosophic set theory quantifies storage risk preferences, while Pythagorean fuzzy decisions capture semantic uncertainties in site suitability. However, challenges persist: ultra-large-scale network optimization faces computational bottlenecks (e.g., nationwide million-variable solving requires distributed computing), high-fidelity geological constraint simulations urgently need machine learning (ML) surrogate acceleration, and cross-regional coordination mechanisms still require exploration. Future methodological innovations must integrate quantum algorithms, digital twins, and other frontier technologies to enable precise deployment of scaled CCUS clusters under global carbon neutrality goals.

4.3. Multiscale models

4.3.1. Global scale

Global CCUS source-sink matching research is progressively revealing the feasibility of transnational carbon mitigation networks. As shown in Fig. 15, Wei et al. [200] integrated data from 3093 emission clusters across 85 countries/regions and 432 storage sites to propose the first global systematic matching scheme. This network could achieve 92 billion tons of CO_2 mitigation, with 64 % via deep aquifer storage and 36 % for CO_2-EOR. Spatial analysis indicates that about 80 % source-sink pairs match within 300 km, primarily concentrated in China, the USA, the EU, Russia, and India. Economic evaluation shows total costs represent merely 0.12 % of the cumulative global GDP, highlighting its scalability potential.

Global regional studies further dissect geographical and geological disparities. The EU GeoCapacity project assessed storage potential in six central-eastern European nations, revealing significant imbalances

Table 6
Mainstream methodology for source-sink matching and pipeline network design in CCUS.

Primary method	Method type	Core feature	Application	References
Mathematical programming	Mixed integer linear programming	Discrete decision variables optimize siting/routing while strictly satisfying capacity constraints.	Pipeline topology design for fixed source-sink scenarios	[194,195]
	Multi-stage MILP	Dynamic multi-period source-sink matching integrating capture-transport-storage scheduling	Full-chain CCUS cluster planning for total cost reduction	[33]
	GIS-enhanced MILP	High-resolution storage potential data enabling pipeline layout optimization	Optimization of injection well deployment with enhanced matching accuracy	[196]
	MILP with pre-optimized networks	Generates candidate pipeline branching networks to prevent a combinatorial explosion	Efficient multi-source/multi-sink matching (SimCCS series)	[180]
	MILP with two-stage linearization	Linearization of nonlinear pipeline/compressor costs with multi-period modeling	Rapid decarbonization network solving under climate targets	[182]
	MILP with a carbon trading mechanism	Allowance trading is integrated into the objective function with emission cap constraints.	Quantifying carbon trading's impact on abatement costs (marginal cost ¥139/t)	[197]
	Multi-objective MINLP (non-linear) with stochastic programming	Piecewise linearization for nonlinear constraints solved via the ε-constraint method	CO_2-EOR network operation control (pump station cost and injection optimization)	[183]
GIS spatial analysis	GIS cost matrix + hierarchical screening	Levelized cost matrix sorting prioritizing the lowest-cost combinations	Rapid national/regional source-sink matching (CAS ITEAM-CCUS)	[198]
	GIS-DEMATEL-ISM integration	Combined geodata and interpretive structural modeling for barrier identification	CCUS deployment potential assessment and policy bottleneck diagnosis	[187]
	GIS-TODIM two-stage model	Integrated resource-risk-environment multidimensional data for suitability computation	Optimal basin and demonstration site selection for CCUS deployment	[199]
	GIS + least-cost objective	Spatial analysis identifies minimum-cost paths and estimates transport routes/costs	Estimating minimum-cost pipeline routes between sources and sinks	[188]
Intelligent optimization algorithms	PFSSPA-MULTIMOORA model	Pythagorean fuzzy sets integrated with MULTIMOORA and priority aggregation operators for uncertainty handling and robust MADM	Site selection evaluation for CGS	[189]
	Graph network theory	Graph-theoretic pipeline optimization incorporating environmental and regulatory constraints	Reducing total pipeline length and abatement costs (U.S. case)	[111]
	Hybrid genetic algorithm-interior point method	Physics-terrain integrated modeling (enhanced fluid dynamics) with global heuristic search optimizing pipe sizing, compressor locations, and cost uncertainties.	Long-distance CO_2 pipeline system design (transport cost minimization)	[190]

between emission intensities and storage suitability: Romania exhibits the highest annual emissions while Hungary has the best theoretical deep aquifer capacity, yet inferior quality constrains practical utilization. Such mismatches necessitate decision support systems integrating geological safety thresholds and economic priorities for transnational pipeline coordination [201]. Innovative technology coupling solutions have also been validated globally, as Castillo et al. [202] evaluated geothermal zones in France, Germany, and the US matched with low-to medium-emission sources, confirming the applicability of combined CO_2-dissolution storage and geothermal recovery, enabling simultaneous CO_2 storage and geothermal extraction to reduce unit abatement costs.

Core challenges for global deployment focus on cross-regional infrastructure coordination and geopolitical alignment. Transnational pipelines require overcoming systemic barriers: technical standard disparities, absent cost-sharing mechanisms, and fragmented regulatory frameworks. Long-distance transport economics are particularly critical, necessitating diversified modes (e.g., shipping and rail) as essential supplements to pipeline networks. Future advancement should prioritize: establishing international storage resource allocation, developing cross-border carbon credit reciprocity rules, and creating standardized full-chain risk governance. These breakthroughs will shift global carbon governance from fragmented deployment to networked collaboration, ultimately building a transnational CCUS ecosystem supporting carbon neutrality.

4.3.2. National scale

Studies on CCUS source-sink matching at the national scale exhibit significant regional differentiation characteristics [203]. To achieve deep decarbonization, retrofitting China's coal-fired power plants is demonstrated as a core pathway, given its status as the world's largest carbon emitter. Comprehensive assessment against the 2 °C target indicates 165 plants, totaling 175 GW in installed capacity, necessitate CCS retrofitting. Suitable geological storage sites are available within a mean radius of 115 km. The total investment required could reach 1.212 trillion USD. Additionally, 23 % of the resultant emission reductions have the potential to yield approximately 37.7 billion USD in revenue through associated EOR technology [204]. China's major sedimentary basins demonstrate immense storage potential, with five basins, including Junggar, listed as priority deployment areas, capable of supporting annual storage capacities of 1.204–1.620 billion tons of CO_2; however, storage efficiency is significantly constrained by transportation distance [205]. Long-term projections indicate that China will need to achieve 0.6 to 1.4 billion tons of CO_2 emission reductions through CCUS technology by 2050; the technology cost is expected to decrease from 310 to 770 CNY/ton in 2030 to 140–410 CNY/ton in 2060, and the source-sink matching potential is fully capable of meeting the demands of the carbon neutrality goal [206].

Emerging economies, however, are overcoming geographical limitations through innovative models [207,208]. As shown in Fig. 16(a), India is pioneering the exploration of cross-border transportation solutions, assessing the economic feasibility of shipping CO_2 to middle eastern oil fields for EOR, providing an alternative decarbonization pathway for the power sector [209]; Its steel industry has planned six cluster projects by matching 15 steel plants with storage reservoirs to achieve annual emission reductions of 108 million tons, with the Maharashtra project utilizing EOR storage technology [210]. As shown in Fig. 16(b), Spain employs a hub-and-cluster model to connect 21 % of the nation's industrial emissions, providing a feasible paradigm for countries lacking oil and gas resources [211]. Fig. 16(c) illustrates Thailand's proposal for six CCS cluster initiatives. These schemes plan to utilize 79 billion tons of saline aquifer storage capacity to capture CO_2 emissions from cement production sites and natural gas power generation plants [212]. Southeast Asian countries rely on clustered

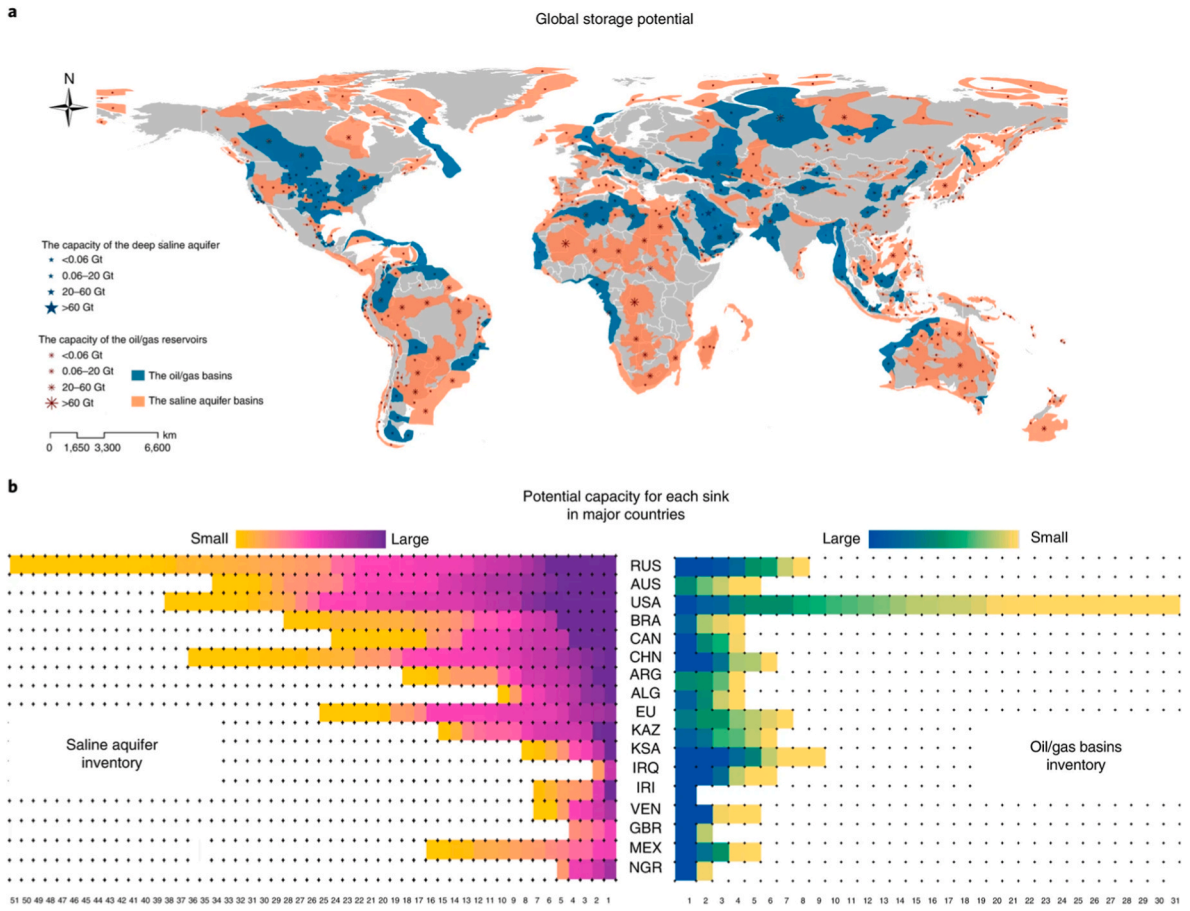


Fig. 15. a. Global distribution of suitable CO_2-EOR reservoirs (blue) with pentagram size indicating storage capacity, and sedimentary basins for aquifer storage (pink) overlaid with stars quantifying CO_2 storage potential. Basins spanning onshore-offshore are displayed as onshore-only, though offshore storage potential remains included. Transnational basin allocations follow proportional territorial area. b. Schematic of per-sink storage potential in top-ranked countries by total capacity. Color intensity in grids denotes storage potential level, with empty grids indicating the absence of corresponding sink types [200].

deployment to optimize cost-effectiveness: As shown in Fig. 16(d), Vietnam's source-sink matching analysis found that establishing four CCUS hubs could achieve annual storage of 14.2 million tons of CO_2 within a 100 km transportation radius, covering 64 % of fixed emission sources [213]. Meanwhile, European and Latin American countries focus on regional collaboration: Croatia has identified three major carbon capture clusters – Adriatic, central, and eastern – prioritizing the development of the eastern region with the shortest source-sink distances [214]; Five CCS-EOR projects identified in Colombia's source-sink matching study have the potential to store 142 million tons of CO_2, addressing 25 % of the petroleum sector's long-term emission mitigation targets [215].

Current challenges and prospects focus on three core issues: Closing the gap between theoretical CO_2 storage capacity and practical geologic availability demands urgent development of enhanced assessment protocols for risk management. Cross-border collaboration mechanisms must overcome the absence of regulatory frameworks and barriers to carbon credit mutual recognition, particularly concerning transnational cost-sharing in shipping and clustered deployment. Economic sustainability relies on coordinated policy innovation to alleviate retrofit cost pressures through carbon tax incentives and the integration of CO_2-EOR revenue cycles. Future large-scale deployment will benefit from continuously declining technology costs and regional pipeline network coordination. Carbon tax incentives, CO_2-EOR revenue recycling, and international allocation mechanisms for storage resources may become critical pathways for emerging economies to break through existing constraints.

4.3.3. Regional scale

Research on regional-scale CCUS source-sink matching highlights context-specific technological pathway innovations. Northeast China optimized a 20-year supply chain deployment using a multi-objective MILP model, minimizing total costs and global warming potential while ensuring emission reduction targets [216]; Further analysis incorporating a real options model indicates that this region possesses the nation's lowest breakeven carbon price (104 CNY/ton in 2024) due to superior storage conditions; policy incentives could increase cement plant CCUS adoption from 13 % to 46 % [217]. The Yellow River basin can achieve resource synergy through CCS-EWR integrated technology: A matching scheme for 236 coal-fired power plants shows annual capture of 739 million tons of CO_2 while supplying 1.314 billion tons of freshwater, with abatement costs controlled below 180 CNY/ton. However, industrial water price incentives have limited effectiveness, necessitating enhanced policy support to jointly address carbon neutrality and water scarcity challenges [218]. China's coastal regions capitalize on 62.4 billion tons of offshore storage capacity in adjacent sea basins, enabling efficient matching where 90 % of emission sources reside within 350 km of sinks. The Huizhou oilfield serves as the designated demonstration site [219]. A dynamic reservoir assessment and multi-well optimization framework developed in Zhejiang province confirms the economic feasibility of mitigating 459 million tons from 154 stationary emission sources, with the optimal CCUS network cost at 51.22 USD/ton [220]. Hebei Province applied a hybrid entropy-AHP evaluation model across three tiers, establishing engineering geology and reservoir characteristics as key site selection factors. The methodology ultimately identified Jing 11 oil well paired with Langfang

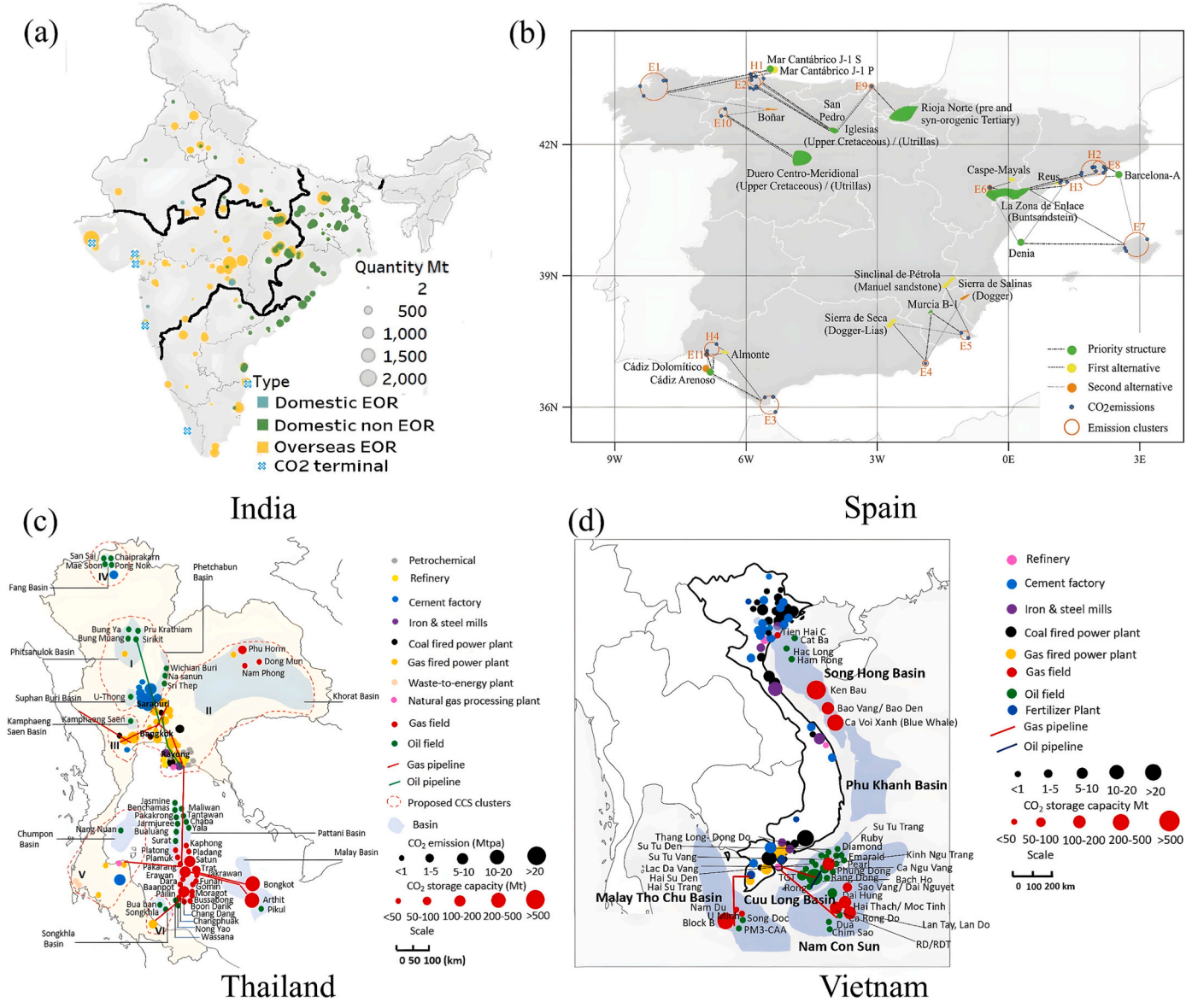


Fig. 16. (a) Distribution of $10/tCO$_2$ revenue across sinks in India, with dark boundaries demarcating power grid regions; (b) Priority source-to-sink clusters in Spain and alternative pipeline configurations; (c) Geospatial distribution of CO_2 sources and sinks in Thailand; (d) Spatial deployment of CO_2 sources and sinks in Vietnam (modified after [209,211–213]).

thermal power plant as the optimal storage solution [221]. Guangdong Province's carbon capture supply chain exhibits a coastal-to-inland diffusion pattern; unit abatement cost rises from 237.81 to 253.51 CNY/ton with increasing emission reduction rates, and a 220 CNY/ton carbon price enables overall profitability [222]. Six carbon mineralization clusters identified via GIS in the Taiwan region offer an annual storage potential of 3 million tons, providing a commercialized synergy solution for power and heavy industries [223].

International case studies deepen the synergy mechanisms between geological resources and industrial layout. Alberta, Canada, matched 157 industrial facilities with 83,470 oil and gas field storage capacities to propose the first CCS corridor spanning 200 km around Edmonton, capable of reducing 38 % of the province's industrial emissions [224]. Research on India's west coast reveals annual emissions of 336 million tons in Maharashtra and Gujarat, with 141 billion tons of storage potential in nearby basins (97.4 % in saline aquifers); four CCS hubs currently in the planning phase have the potential to cut stationary emissions by half [225]. The Gulf of Mexico developed a multi-dimensional decision framework integrating geological features, logistics networks, and risk factors; after systematically evaluating 2559 sites on the outer continental shelf, the Louisiana coast was identified as the optimal region for EOR storage due to highly permeable reservoirs and dense infrastructure [226].

Core challenges in current regional practices include significant disparities in breakeven carbon prices (e.g., 104 CNY/ton in NE China vs. 220 CNY/ton in Guangdong) and insufficient development of derivative value (e.g., underutilization of water recovery benefits in the Yellow River basin). Future breakthrough pathways should focus on three key aspects: deepening revenue mechanisms for hybrid technologies, such as water-carbon synergy and mineralization storage; establishing regionally differentiated carbon price compensation policies to balance economic feasibility; and promoting standardized geological risk assessment to accelerate the transition of regional CCUS from single-point demonstrations to networked deployment.

4.3.4. *Industrial scale*

Research on industrial-scale CCUS source-sink matching reveals differentiated decarbonization pathways for key industrial sectors. As

shown in Table 7, a systematic review is provided of sector-specific studies across coal power, cement, steel, refining, waste incineration, and integrated industries—highlighting research focus, national contexts, and key findings. The table synthesizes critical technical-economic assessments and infrastructure requirements, facilitating cross-sectoral and cross-regional comparisons. It serves as a consolidated reference for identifying cost-effective deployment patterns, policy-sensitive thresholds, and scalable storage-transport configurations, thereby supporting strategic planning and large-scale CCUS implementation in hard-to-abate industries.

Chinese research identifies coal-fired power plants as key emission reduction targets, revealing that more than 70 % achieve cost-competitiveness with natural gas facilities post-carbon capture implementation. Furthermore, 22–58 % demonstrate economic viability comparable to onshore wind power, indicating significant feasibility under policy incentives [186]. Further analysis of 584 coal plants shows an abatement potential of 5.46–38.48 billion tons during 2020–2060, with 300 km as the optimal transport distance threshold. However, declining operating hours significantly increase capture costs, necessitating subsidy mechanisms to compensate for losses [227]. Innovative technology synergies like CCS-EWR could achieve annual reductions of 1.32 billion tons while supplying 38.48 billion tons of freshwater, covering 70.7 % of units under a 400 CNY/ton carbon price [228]. The cement industry faces challenges in balancing cost and scale: achieving 20 %–60 % emission reductions requires retrofitting 229–627 plants at an economic cost of 79.4–269.5 billion CNY, though Northeast and Northwest China are prioritized due to superior storage conditions; selected 603 cement plants are projected to reduce emissions by 1.85 billion tons during 2030–2060 [229]. The steel industry highlights a spatial mismatch conflict—111 blast furnace-basic oxygen furnace plants have an annual abatement potential of 420 million tons, but the separation of eastern emission sources and western storage sites causes transport costs to rise to 83–118 USD/ton as capture targets increase (30 %/60 %/90 %), with current carbon prices supporting immediate investment in only 14 projects [230].

Cross-sectoral collaboration and exploration of emerging fields expand application scenarios. Integrated modeling of China's coal power, steel, and cement sectors demonstrates that deploying carbon capture across 658 power plants, 101 steel mills, and 357 cement facilities by 2045 could yield annual CO_2 reductions of 1.444 billion tons. This transformation would necessitate $31.247 billion USD in investment, with Jinan and Shenyang designated as cluster cores [231]. A land-sea integration scheme connects 1186 emission sources to 329 storage sites via 80,700 km of shared pipelines, quantifying sectoral storage potential for coal power (175 million tons/year), steel (77 million tons), cement (56 million tons), and coal chemical (23 million tons) [232]. India focuses on the refining sector, identifying four source-sink matching schemes and analyzing barriers to first-time EOR implementation [233]; Canada's oil sands sector designed a hub network linking 10 emission sources, capable of storing 190 million tons of CO_2 but requiring a 4 billion CAD investment [234]. Emerging sectors like waste-to-energy have annual emissions of 318.5 million tons, concentrated in eastern coastal China; 996 plants could match storage sites within 800 km, with 147 plants having abatement costs as low as 91.1 CNY/ton, but carbon-water synergy management conflicts need resolution [235].

The significant research focus on CCUS source-sink matching at the industrial scale in China is driven by multiple urgent factors. First, as the world's largest carbon emitter, China faces tremendous pressure to achieve its ambitious "Dual Carbon" goals (carbon peak by 2030 and carbon neutrality by 2060). The industrial sector—especially coal power, steel, and cement—accounts for a significant portion of the country's emissions, making it a critical target for decarbonization. Second, China's energy structure is dominated by coal, and its heavy industries are concentrated in eastern regions, while large-scale geological storage sites are mainly located in the west, creating a

Table 7
Industrial-scale CCUS deployment patterns: technical-economic assessments and infrastructure requirements in key industries.

Industry	Country	Core research focus	Key findings	References
Coal power	China	Economic feasibility of CCUS retrofitting for coal-fired power plants	Post-capture, >70 % of coal plants become cost-competitive with gas plants; 22–58 % rival onshore wind economically	[186]
	China	Optimization of CCUS supply chains for coal power plants	Offshore storage offers greater advantages for southern coastal provinces, enabling 4.2 billion tons of carbon storage within a 250 km transport radius; unit cost increases by 26 % when annual deployment scales to 1.2 billion tons	[236]
	China	Lifecycle emission reduction potential of coal power plants	584 coal power plants possess 5.46–38.48 Gt abatement potential between 2020 and 2060, with 300 km identified as the optimal upper limit for transport distance	[227]
	China	Impact of operating hours on costs	Declining operating hours significantly increase carbon capture costs but reduce transport costs, necessitating subsidies to compensate for power generation losses	[237]
	China	Synergistic effects of CCS-EWR technology	Retrofitting coal power plants with CCS-EWR enables annual CO_2 reductions of 1.32 Gt while supplying 38.48 Gt of freshwater. At a carbon price of 400 CNY/ton, this approach renders 70.7 % of units economically viable.	[228]
Cement	China	Source-sink matching model for optimized deployment	Deploying CCUS at 229–627 cement plants is required to achieve 20 %–60 % emission reduction targets, resulting in an economic cost of 794–2695 billion CNY	[238]
	China	Screening of cement plants for CCUS retrofitting	603 feasible cement plants were screened; they are projected to achieve 1.85	[229]

(continued on next page)

Table 7 (*continued*)

Industry	Country	Core research focus	Key findings	References
			billion tons of CO_2 reduction between 2030 and 2060, with northeast and northwest China as priority implementation regions	
Steel	China	Emission reduction potential of blast furnace-basic oxygen furnace steel plants	111 steel plants have an annual emission reduction potential of 420 million tons through full-chain CCUS projects; current carbon prices support immediate investment in only 14 projects	[230]
	China	Analysis of east-west spatial mismatch	Significant spatial mismatch exists between CO_2 emission sources (concentrated in the east) and storage sites (concentrated in the west); transport costs dominate the total cost as capture targets increase to 30 %/60 %/90 %	[34]
Refining	India	Matching refineries with EOR	Four source-sink matching schemes were identified, and challenges for the first-time implementation of CCUS technology were analyzed	[233]
Waste incineration	China	Source-sink matching optimization for incineration plants	318.5 million tons/year of waste incineration CO_2 emissions are mainly concentrated in eastern coastal China; storage sites can be matched within 800 km for 996 plants	[235]
Oil sands	Canada	CCUS hub design and financing	Designed a pipeline network connecting 10 emission sources, capable of safely storing 190 million tons of CO_2, requiring a 4 billion CAD investment	[234]
Integrated industries	China	Joint optimization of coal power + cement + steel	To achieve an annual reduction target of 1.444 billion tons, CCUS retrofitting is required for 658 power plants, 101 steel plants, and 357 cement plants by 2045	[231]
	China	Land-sea integration for coal power + steel + cement + coal chemical	Connects 1186 emission sources to 329 storage sites via an 80,700 km shared pipeline network, establishing a complete geospatial connectivity pattern	[232]
	India	Assessment of power plants and industrial facilities	630 power plants and industrial facilities emit 170 million tons of CO_2 annually; storage capacity can meet demand for 237 years	[239]

pronounced spatial mismatch that necessitates optimized source-sink matching. Moreover, policy promotion and financial support from the government, along with the need to ensure energy security while reducing emissions, have further accelerated research and large-scale assessments in this field. These factors collectively explain China's leading role in industrial-scale CCUS research and deployment.

Current challenges and prospects: Core bottlenecks for sectoral deployment include transport costs inflated by the steel sector's east-west spatial mismatch, carbon-water synergy conflicts in waste incineration, and massive capital requirements for multi-sector retrofits. Future breakthroughs should focus on three aspects: (1) developing shared pipeline networks to reduce mismatch costs (2) establishing policy portfolios (e.g., carbon price-subsidy-water rights trading); to resolve sectoral economic barriers; (3) formulating differentiated standards for geological storage utilization. China's dominance in sectoral research stems from the urgency of heavy industry decarbonization under its carbon neutrality goal—industrial scale (670 GW coal power capacity, 53.3 % global steel output) drives large-scale assessment needs, while policy-driven emission reduction targets further accelerate research on optimizing technological pathways. International experience indicates that promoting the "clustered retrofit + geological suitability" model will become a key pathway for deep industrial decarbonization.

5. Challenges and prospects

This paper systematically reviews the entire technical pathway of CCUS source-sink matching, revealing that its large-scale development still faces four interrelated systemic constraints (Fig. 17). First, there are multi-level disconnections within the technology-economy-policy system; simplified cost estimation, lack of precision in policy incentives, and barriers to cross-sectoral coupling impede commercialization. Second, insufficient data accuracy and scale mismatch between carbon sources and sinks lead to deviations between matching results and actual demands. Third, rigid methods for transport network optimization and insufficient cluster collaboration limit the system's flexibility and economic viability. Fourth, the disconnection between intelligent algorithms and physical mechanisms, coupled with the lack of dynamic evolution characterization, makes it difficult to support precision optimization in complex scenarios. These challenges are intertwined and collectively constitute the core obstacles to source-sink matching.

5.1. Synergy bottlenecks in techno-economic policies

The CCUS source-sink matching techno-economic-policy system exhibits multi-tiered disconnects, primarily manifested in three

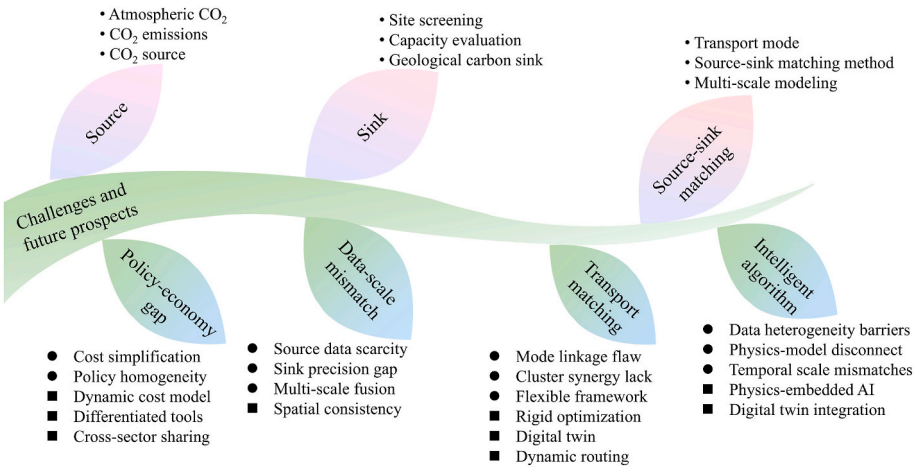


Fig. 17. Challenges and prospects for CCUS source-sink matching.

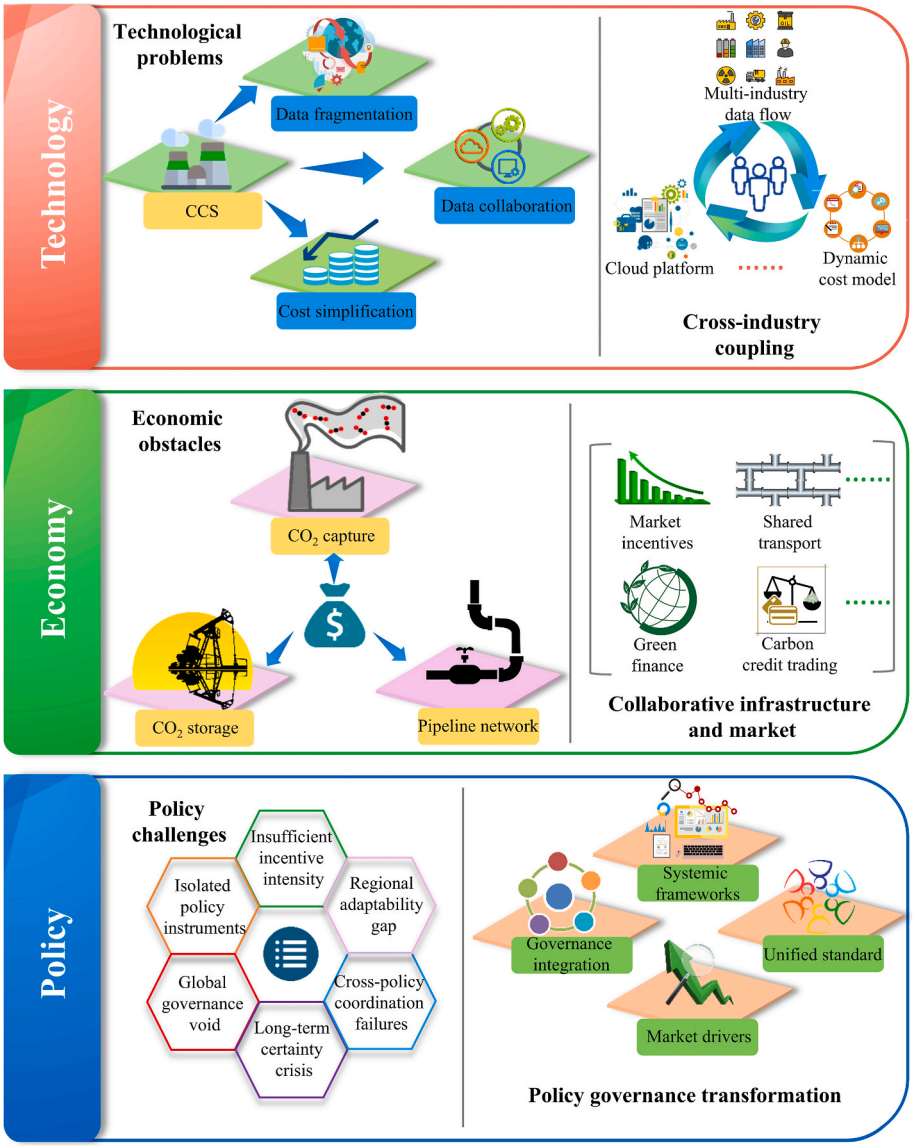


Fig. 18. The technology-economy-policy challenges and prospects of CCUS.

dimensions (Fig. 18). First, oversimplified cost estimation constrains commercial viability: Existing models often reduce full-chain CCUS costs (e.g., capture, transport, and storage/utilization) to linear functions, failing to capture scale effects from clustered deployment or dynamic cost fluctuations driven by technological learning. For instance, pipeline network sharing among large point sources can reduce costs, yet such benefits are rarely incorporated into economic assessments [232]. Concurrently, inadequate granularity in cost breakdowns—insufficient delineation between fixed costs (e.g., capture equipment installation and pipeline construction) and variable costs (e.g., CO_2 transportation and injection maintenance)—hinders precise identification of cost optimization opportunities. Second, policy incentives lack targeting: Current carbon pricing mechanisms (e.g., carbon taxes and subsidies) predominantly apply uniform regional standards, overlooking spatial disparities in source-sink matching difficulty—e.g., high transport costs for cross-regional matching in Western China necessitate differentiated price signals [34]. Furthermore, poor articulation between policies and market mechanisms results in carbon market price signals failing to effectively guide optimal source-sink matching configurations, leading to mismatches between policy incentives and actual abatement potential in certain regions [240]. Third, economic barriers impede cross-sectoral coupling: While biomass-coal power coupling or renewable energy-industrial park integration can reduce capture costs, fragmented CO_2 flow data across multiple industries hinders model-based quantification of synergistic benefits. For collaborative capture between coal power and cement sectors, potential benefits from shared energy consumption and cost allocation remain unquantified due to the absence of unified data standards and accounting methodologies.

Future pathways include: (1) Developing dynamic cost models integrating cluster effects and technology learning curves to incorporate economies of scale and technological progress into cost accounting, enhancing prediction accuracy (2) Designing differentiated policy instruments (e.g., transport-distance-based carbon tax rebates); to mitigate spatial mismatches, while strengthening policy-carbon market linkages for more precise price signal-driven resource allocation; (3) Establishing cross-sectoral benefit-sharing mechanisms—such as reinvesting CO_2-EOR revenues into high-cost industrial capture—and advancing unified cross-industry data platforms to support synergistic benefit quantification, fostering deeper inter-sectoral collaboration in source-sink matching.

5.2. Data accuracy and scale mismatch

The accuracy of source-sink matching is constrained by significant gaps in data precision and scale integration, an issue prominently manifested in multidimensional assessments of both CO_2 sources and sinks. For CO_2 sources, granular emission data at process-level (e.g., specific blast furnaces in steel plants and cement kilns) and grid-level (1 km $\times$ 1 km) are severely lacking, forcing models to rely on sector-average emission factors for estimation, which may lead to overestimation or underestimation of abatement potential. Early models such as SimCCS utilized only regionally aggregated emission data, failing to capture distribution patterns of localized high-emission hotspots; Even in upgraded models, Wei et al. [200] refined data sources to national or continental-scale industrial clusters but still lacked dynamic integration of real-time monitoring data (e.g., online flue gas analysis), rendering them unable to adapt to emission variations caused by production load fluctuations.

For CO_2 sinks, precision disparities across assessment scales further exacerbate matching challenges: Basin-scale storage potential is often macro-estimated based on geological formations, failing to directly inform site-level injection well design; Moreover, microscale uncertainties in key reservoir parameters (e.g., porosity, permeability, and maximum injection rates) propagate through modeling layers, introducing systematic errors into source-sink matching outcomes [241]. For instance, Fan et al. [205] incorporated carbon sink data across 3853 high-resolution grids, yet the grid scale remained insufficient to accurately characterize the impacts of small-scale faults on CO_2 migration, potentially overestimating actual storage security [227].

Additionally, the issue of data scarcity represents a global challenge that varies significantly across countries. In developed nations such as those in North America and Europe, relatively comprehensive geological databases and established monitoring systems provide a stronger foundation for source-sink matching assessments. However, many emerging economies and developing regions still suffer from insufficient publicly available high-resolution data on both emission sources and storage formations. This disparity not only hinders accurate regional capacity evaluations but also limits the feasibility of cross-border carbon management strategies. The lack of standardized data collection protocols and sharing mechanisms further exacerbates these challenges, creating barriers to global CCUS deployment and international collaboration in climate mitigation efforts.

Future perspectives include: (1) Establishing a multi-scale data fusion platform to vertically integrate satellite remote sensing (grid-level), enterprise online monitoring (point-source-level), and production process data (process-level), generating continuous emission-storage datasets via spatiotemporal interpolation algorithms; (2) Developing probabilistic carbon sink assessment models incorporating Monte Carlo simulations to quantify reservoir parameter uncertainties, translating them into risk constraints for matching optimization; (3) Applying ML algorithms to upscale site-specific data (e.g., core tests and well testing) to basin-level, enhancing spatial matching consistency through geological model calibration.

5.3. Limitations in transport and matching mechanisms

The limitations of transport network optimization and source-sink matching algorithms primarily manifest in rigid methodologies and insufficient coordination, significantly constraining the flexibility and cost-effectiveness of CCUS systems [242]. First, rigid optimization methods struggle to adapt to complex dynamic scenarios. Although mainstream MILP models serve as the core methodology for most matching frameworks, their reliance on predefined pipeline routes and static parameters hinders responsiveness to dynamic adjustments during actual operations [243,244]. For instance, models cannot efficiently replan transport routes in real-time during temporary CO_2 supply disruptions (e.g., equipment maintenance) or demand fluctuations (e.g., EOR project injection rate adjustments); Cost allocation and scheduling logic for multimodal transport interfaces (pipeline-vessel-truck) are often oversimplified; while MILP models can estimate terrain impacts on pipeline construction, they fail to optimize port handling efficiency at maritime hubs or vessel scheduling cycles, leading to underestimated total costs for transregional matching (e.g., European industrial sources to Norwegian North Sea storage) [173]. Moreover, early models rarely incorporated critical constraints like injection capacity or pipe thickness into optimization (e.g., ignoring economies of scale in pipeline costs). While recent models show improvements, they still struggle to balance computational complexity with practical engineering feasibility.

Second, cluster-scale multimodal coordination is underdeveloped in CCUS. Most existing models focus on single pathways—e.g., CO_2-EOR in Xinjiang Oilfield or saline aquifer storage in Ordos Basin, ignoring coupled optimization of multiple modes [245,246]. Integrating CO_2-EOR, CO_2-ECBM, and CO_2-EWR could share compression stations, pipelines, and monitoring systems, cutting per-unit energy use while balancing short-term profits (from EOR/ECBM) and long-term sequestration goals. Coal-to-hydrogen coupled with CCUS via shared pipelines also reduces transport costs, yet static domestic source-sink models lack such logic, leading to fragmented cluster operations, higher overall energy consumption, and missed resource-sharing cost cuts.

Moreover, synergies between CO_2 chemical utilization (e.g., methanol production) and geological storage, plus new energies, are excluded from cluster frameworks due to no cost-benefit models. In coastal

clusters, part of captured CO_2 can serve nearshore chemical synthesis, with the rest sent to offshore aquifers via shared tankers. Synergies with CO_2 storage and BECCS could boost circularity—reservoirs absorb geothermal energy, biomass supplies low-carbon heat for capture. But current designs ignore these, underusing resource circularity and limiting cluster emission reduction efficiency.

Future breakthroughs include: (1) Developing flexible optimization frameworks, such as embedding real-time constraints (e.g., supply-demand fluctuations and weather impacts) into MILP models, combined with ML algorithms for dynamic multimodal transport routing to enhance resilience in transregional matching; (2) Constructing "storage-utilization" coupled cluster models, for example, linking cement kiln CO_2 mineralization with adjacent saline aquifer storage, quantifying co-benefits via material flow analysis; (3) Leveraging digital twin technology to integrate geological, meteorological, and operational data, simulating transport network resilience under seismic or storm risks to support decision-making for intercontinental source-sink matching.

5.4. Intelligent algorithms and data-driven approaches

The integration of ML algorithms with source-sink matching for CCUS represents an emerging methodology that demonstrates significant advantages in handling high-dimensional nonlinear systems, multi-source heterogeneous data fusion, and dynamic optimization. Compared to traditional methods like MILP and GIS spatial analysis, ML approaches exhibit distinct differences while also possessing complementary characteristics: MILP excels at solving structured optimization problems, possessing a rigorous mathematical foundation, particularly for pipeline network topology design and cost minimization; however, its reliance on pre-defined constraints and linear assumptions makes it inflexible in handling real-time data fluctuations and multi-physics coupling problems. GIS spatial analysis offers prominent advantages in macroscopic spatial matching and visualization, enabling rapid identification of the geographic distribution characteristics of sources and sinks; nevertheless, its static data processing approach fails to capture uncertainties in geological parameters and dynamic evolutionary processes.

In contrast, ML methods leverage data-driven pattern recognition and predictive capabilities to extract implicit patterns from massive heterogeneous data, demonstrating strong potential, particularly in handling unstructured geological data (e.g., seismic interpretation results and core experimental data), multi-scale parameter inversion (e.g., porosity and permeability estimation from basin to reservoir scale), and dynamic risk early warning (e.g., CO_2 leakage probability prediction) [247]. For example, algorithms such as Random Forest and XGBoost can achieve high-precision storage potential predictions in specific geological scenarios, while deep learning models are increasingly being applied in complex geological body identification and CO_2 migration simulation, effectively compensating for the limitations of traditional methods [248].

Despite demonstrating unique value in CCUS source-sink matching, the application of ML still faces three core bottlenecks: firstly, the heterogeneity of geological data limits model generalizability; divergent geological backgrounds across regions (e.g., sedimentary environments, rock mineral composition) lead to a lack of unified standards for unstructured geological data, and insufficient sample space coverage of key parameters (e.g., caprock breakthrough pressure, reservoir permeability) causes a significant drop in prediction accuracy when models are applied across regions [249]. Secondly, the disconnection between physical mechanisms and data models reduces prediction reliability; models like deep neural networks are often regarded as "black boxes," making it difficult to internalize multi-physics coupling processes (e.g., CO_2-water-rock interactions and pore pressure transmission), leading to prediction biases in tectonically active zones or heterogeneous reservoirs [250]. Thirdly, existing models lack the capability to represent dynamic evolution, failing to establish connections between long-term geological evolution (e.g., mineral carbonation reaction rates) and short-term monitoring indicators (e.g., injection well pressure changes), resulting in a disconnect between storage capacity assessment and leakage risk on a temporal scale, which hampers long-term source-sink matching optimization [251].

Future breakthrough pathways include: embedding geo-mechanical constitutive equations as physical constraints into deep neural networks, developing surrogate models capable of resolving multi-field coupling, thereby enhancing the cross-regional applicability of site screening; integrating short-term scheduling and long-term evolution models based on a digital twin, utilizing filtering assimilation to update parameters in real-time, to build a full-chain time-varying decision-making system; combining advanced algorithms with edge computing, designing a parallel engine to synchronously handle objectives of pipeline cost minimization, risk controllability, and multi-resource synergies, generating multi-objective optimization schemes. Ultimately, this will drive the transition of source-sink matching from "static style optimization" to "dynamic system collaboration", supporting the global clustered deployment of CCUS.

6. Summary, and recommendations for future work

This paper focuses on the critical role of CCUS technology in addressing the global climate crisis. It systematically reviews recent advances in source-sink matching. By establishing a three-dimensional "source-sink-matching" analytical framework that integrates multi-scale elements—global spatiotemporal distribution of emission sources, geological storage capacity assessment, and pipeline network optimization. It reveals key challenges such as spatial source-sink mismatch and the dominance of transportation costs. It emphasizes the irreplaceability of optimized matching mechanisms for reducing abatement costs and achieving the Paris Agreement's 1.5 °C target. Recommendations for future work are outlined as follows.

(1) Current matching models predominantly rely on static parameters, making them poorly adaptable to dynamic emission fluctuations and storage performance evolution. Future work could integrate real-time emission monitoring data, reservoir pressure response simulations, and carbon market dynamics to develop multi-objective dynamic programming algorithms, enabling collaborative optimization of sources, sinks, and pipeline networks.
(2) Existing research focuses predominantly on large stationary sources (e.g., oil refinery, cement factories, and coal-fired power plants), overlooking the potential of distributed emission sources. Future efforts could develop capture technologies for small-to-medium mobile sources (e.g., shipping and aviation), explore regional carbon hub cluster models, and enhance coverage across all industrial sectors.
(3) Quantitative tools are lacking for assessing public perception and policy risks in site screening. Future work could introduce GIS-based socio-spatial analysis to develop a three-dimensional risk assessment system encompassing "geological suitability-community acceptance-regulatory compliance."
(4) Geopolitical barriers exist between global emitting nations and storage-resource-endowed countries. Future efforts could design a transnational framework for carbon credit allocation and responsibility sharing, to activate cross-border markets for storage resources.
(5) Current CCUS source-sink matching models over-rely on conventional mathematical programming methods. It is recommended to develop new highly efficient algorithms and ML-driven intelligent optimization frameworks, utilizing data assimilation techniques to jointly analyze dynamic characteristics of emission sources, storage evolution patterns, and pipeline

regulation capabilities, to enhance matching efficiency and cross-scenario adaptability systematically.

(6) It is advised to integrate well-log and seismic data with other multi-source heterogeneous geotechnical data to establish high-precision intelligent site screening and capacity assessment methods for diverse utilization scenarios including EOR, EGR, EWR, ECBM, and basalt storage.

(7) Future work should explore integrated planning models for CCUS coupled with CCU, renewable energy (e.g., green electricity for hydrogen production/heating), and hydrogen pipeline networks, to unlock integrated emission reduction potential.

(8) Future efforts could establish a multi-dimensional evaluation and visual interpretation framework spanning global to enterprise scales, covering engineering parameters, economic costs, environmental impacts, policy constraints, and carbon neutrality pathways.

(9) Future research should develop dynamic carbon footprint accounting models at the source-sink node level with high spatio-temporal resolution, to support more granular cost-benefit analysis and policy design.

(10) There is an urgent need to establish a unified spatiotemporal reference data integration platform based on GIS big data, integrating multidimensional information encompassing geology, environment, infrastructure, and economic policies, to achieve visual dynamic management of matching elements.

Abbreviations

CCUS	Carbon capture, utilization, and storage
CO_2	Carbon dioxide
IPCC	Intergovernmental Panel on Climate Change
CSLF	Carbon Sequestration Leadership Forum
US-DOE	United States Department of Energy
CH_4	Methane
N_2O	Nitrous oxide
GWP	Global Warming Potential
IEA	International Energy Agency
CCS	Carbon capture and geological storage
GHG	Greenhouse gas
CGS	CO_2 geological storage
GIS	Geographic information system
USGS	United States Geological Survey
GCCSI	Global CCS Institute
NACSA	North American Carbon Sequestration Atlas
EU	European Union
CO_2-EOR	CO_2-enhanced oil recovery
MMP	Minimum miscibility pressure
CO_2-ILU	CO_2 storage coupled with in-situ uranium leaching
CO_2-ESWR	CO_2-enhanced saline water recovery
CO_2-ECBM	CO_2-enhanced coalbed methane
CO_2-EGR	CO_2-enhanced gas recovery
CO_2-EGS	CO_2-enhanced geothermal systems
CO_2-ESGR	CO_2-enhanced shale gas recovery
LNG	Liquefied natural gas
MILP	Mixed-integer linear programming
MINLP	Mixed-integer nonlinear programming
LCPA	Least-cost path analysis
NETs	Negative emission technologies
AHP	Analytic hierarchy process
ML	Machine learning

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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